

SoK: The Deadly Race

When Graceful Recovery Makes Coercion-Resistant Custody Worse Than Useless

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Abstract. Physical-coercion (“wrench”) attacks defeat custody schemes that key management and cryptography treat as out of scope, and the countermeasures the industry has shipped against them have never been assessed against a common bar. We supply one, and use it to systematize the design space. We give a formal model of coercion against a secret-holder and prove that a large class of “graceful” recovery designs — those that leave a device seizure holding a *feasible-but-slower* path to the secret — are strictly *worse than useless*: they manufacture a material incentive to *assassinate* the victim. We name this the **Deadly Race**, derive it from a single impossibility (one cannot prove one has forgotten), and distill from it a design bar — the **No-Gray-Area Criterion** — with a *reachability* dichotomy that partitions every way of clearing it, and a trichotomy that enumerates the mechanisms available to a design that tries. Against that taxonomy we classify what is actually deployed: time-lock-rescue vaults clear the bar honestly, by putting the winning racer beyond the attacker’s reach, at the cost of a trusted third party; the standard industry control set partitions cleanly into the configurations that clear it and those that are worse than their own absence; and the most widely shipped consumer countermeasure, the self-destruct (“wallet-erase”) PIN, clears nothing — the vendor’s own documentation presupposes a surviving backup, so the erasure removes the device path while leaving intact the path that runs through the holder. Two commonly held assumptions do not survive the exercise: that discretion about ownership is a security mechanism, and that a duress PIN is a coercion defense. A construction, the **Time-Gated Perceptual Oracle**, shows the bar is reachable by the other route with custody kept *individual*. Coding of 352 incidents from the physical-attack registry grounds the model’s stipulations, attests the predicted terminal outcome, and bounds the delegated route empirically. The framework fixes a bar; it does not claim this construction is the unique way to clear it.

Keywords: Bitcoin self-custody · physical-coercion (wrench) attacks · coercion resistance · Kerckhoffs’s principle · threat modeling.

1 Introduction

Coercion is the unmodeled attack. Standard threat models — Dolev–Yao on the wire [1], IND-CCA for encryption, forward secrecy for sessions — all assume the

adversary cannot compel the *holder* of a secret. For bearer assets that assumption is empirically false: the physical attack registry maintained by Lopp [2], and the systematic study of the phenomenon in the financial-technologies literature [3], document a growing series of incidents in which a Bitcoin holder is physically threatened with a “\$5 wrench” until they surrender their keys.

Cryptographic hardness does not help here. A wrench attacker does not break the cipher; they compel the person who can operate it. Worse, the standard defense offered to holders — *obscurity* (“never tell anyone you own Bitcoin”) — is not a security mechanism at all in the sense of Kerckhoffs [4]: it is a bet that the attacker lacks knowledge the defender has, and it degrades precisely as adoption spreads that knowledge. A 2026 case already shows the degradation in miniature: a streamer who denied holding any Bitcoin was tortured for hours regardless, his denial simply disbelieved [5] (we return to it in §5.5). Nor is the leak channel usually the holder’s own mouth: threat-intelligence reporting on the 2026 wave attributes victim selection to *leaked databases, tax records and exchange data*, alongside insider sales of customer lists and dark-web recruitment for employee access [6] — records held by third parties, which the holder can neither audit nor revoke once created.¹ This “keep a low profile, never disclose” counsel is nonetheless the community’s entrenched consensus, actively promoted by prominent voices (cataloged in the accompanying economic analysis [7]); and it carries a measurable price — that analysis finds the severity-adjusted actuarial cost of individual Bitcoin self-custody already *exceeds* the cost priced into insurance for comparable portable valuables such as jewellery.

This paper models coercion directly, and uses the model to systematize a design space built without one. It is worth saying at the outset what kind of result that yields, since it is not the kind a cryptographic reading looks for: a *decision-theoretic* impossibility with empirical grounding, addressed to the designers of custody systems, with no primitive proposed, no reduction given and nothing cryptanalyzed. The Deadly-Race Lemma holds *whatever* the underlying cryptography, because it lives in the incentive layer that threat models omit — which is why “the crypto is strong” is no answer to it. Our contributions are:

1. a **formal coercion model** (§2) with an explicit adversary, a Kerckhoffs premise, and an engineering “feasibility line” that turns asymptotic security into concrete parameter bounds;
2. the **Deadly-Race impossibility chain** (§3): a classical impossibility (you cannot prove you forgot) that escalates, step by step, into an incentive to murder the victim;

¹ The counsel’s strongest form answers that such records need never exist: acquire only without identity verification, and keep no reportable trail. We grant that this reduces exposure, but three things keep it from being a security *mechanism*: exposure ratchets, a leak being impossible to un-leak; the compliant holder cannot lawfully follow it, since where holdings are reportable, leaving no fiscal record is non-declaration; and, decisively here, Assumption 1 puts any ignorance-contingent defense out of scope by construction. Appendix G gives the argument in full.

3. the **No-Gray-Area Criterion** (§3), the design bar — established as a theorem, not a stipulation — with a *reachability* dichotomy partitioning every clearance route and a *trichotomy* enumerating the mechanisms that can take one;
4. a **systematization of what is deployed** (§5) against that taxonomy: two time-lock-rescue vaults, the standard industry control set, and the self-destruct PIN class shipped by three vendors, each placed by which route it takes or why it takes none — which refutes, from the vendors’ own documentation, the assumption that a duress PIN is a coercion defense;
5. an **empirical grounding** (§5.5) in a coding of 352 physical-attack incidents, calibrating the stipulations, attesting the predicted terminal outcome, and bounding the delegated route; and
6. a **construction** (§4), the TGPO orbit, establishing that the remaining route is reachable with custody kept individual.

We call the central results the **Deadly-Race Lemma** and the **No-Gray-Area Criterion**, after their first statement in the protocol’s public design record [8]. Supporting material is gathered in the appendices; the paper is intended to be read without them.

2 Model

2.1 Principals and capabilities

The **holder** $\mathcal{V}$ possesses whatever is needed to reach a master secret K gating an asset worth W ; the **coercer** $\mathcal{A}$ seeks K ; optional **delegates** may hold partial capability on $\mathcal{V}$ ’s behalf. Appendix E tabulates every symbol.

Assumption 1 (Kerckhoffs, premise 0). *$\mathcal{A}$ knows the protocol in full: all algorithms, all source, $\mathcal{V}$ ’s setup topology, and every in-scope parameter. Only the protocol secrets are hidden. Every result below is proved against such an $\mathcal{A}$; security by obscurity is assumed to contribute nothing.*

Assumption 2 (The Wrench–Dolev–Yao (WDY) adversary). *The Wrench–Dolev–Yao (WDY) adversary inherits the Dolev–Yao [1] intruder on the wire but departs from it twice. It is computational, not symbolic: perfect cryptography is replaced by the feasibility line (Stipulation 1), so “cannot break the cipher” becomes “cannot exceed its work budget”. And it is extended past the wire with the physical capabilities Dolev–Yao omits — device seizure and a bounded coercion oracle over the holder: the wrench the name records. WDY is fixed by:*

D0. Kerckhoffs. *$\mathcal{A}$ knows the protocol, $\mathcal{V}$ ’s setup topology, and every in-scope parameter (Assumption 1).*

D1. Bounded compute. *$\mathcal{A}$ is bounded by the feasibility line (Stipulation 1): 2^{32} cheap evaluations feasible, plain 2^{64} cheap at the border (the softest number), 64^* and 2^{96} cheap both infeasible. This is what replaces “unbounded”.*

- D2. **Frequency ceiling.** $\mathcal{A}$'s hardware evaluates a computation only boundedly faster than $\mathcal{V}$'s, so an entropy-preserving hash with enough non-parallelizable steps (H^*) imposes an arbitrarily long per-evaluation latency on $\mathcal{A}$ — at a proportionally longer one on $\mathcal{V}$. This is what makes the time-gate meaningful: $\mathcal{A}$ can neither parallelize nor out-clock its way past a sequential memory-hard chain.*
- D3. **Complete subjugation.** This is an information-security model, not a study of violence against resistance, so we do not model $\mathcal{A}$'s coercive power. To keep the argument a fortiori we assume that from the onset of the attack until release $\mathcal{A}$ meets no resistance whatsoever and enjoys $\mathcal{V}$'s full cooperation; the sole qualification, feigned cooperation in the residual window, is of residual importance (§4.4).*
- D4. **Co-located device seizure.** All of $\mathcal{V}$'s devices at the attack site are immediately seized — vault blobs, time-lock blobs, commitment scripts, review schedules. A single WDY encounter has one site; devices and principals at other sites lie beyond its reach within the bounded window — the opening a geographic defense exploits (§3.6).*
- D5. **Explicit-secret seizure.** All of $\mathcal{V}$'s explicit (non-tacit) authenticating secrets are immediately seized.*
- D6. **No tacit seizure.** $\mathcal{V}$'s tacit entropy-carrying secrets — the points — are captured iff they are available in transmissible form at the moment of attack (materialized, and not wiped before $\mathcal{V}$ is subdued): exactly when the tacitness fails. When it holds they are not, and by D3 that is a genuine incapacity to transmit, not unwillingness — even a fully cooperating $\mathcal{V}$ cannot hand over what is recognizable but non-transmissible.*
- D7. **No perceptual-parameter seizure.** Likewise $\mathcal{V}$'s perceptual-oracle parameters — the deep fractals — are seized iff available at the moment of attack: they are regenerated only by walking the memory-hard chain, so a willing $\mathcal{V}$ has nothing to dictate. This clause and D6 are the operational restatement, in the adversary model, of the tacit-knowledge premise (Assumption 3).*
- D8. **No remote-delegate coercion.** A remote delegate — a party holding rescue capability beyond the attack site (D4) — is past $\mathcal{A}$'s remote coercive reach: from the encounter with $\mathcal{V}$, $\mathcal{A}$ cannot compel it to surrender or act on its state. Without this a delegated defense is vacuous. Real delegates span a spectrum of coercion-compliance; collapsing it to “cannot be reached” is a deliberate simplification, at the cost of reality's shades.*

The only channel to the tacit material withheld by D6–D7 is a **bounded coercion oracle**: $\mathcal{A}$ extracts it through the cooperating $\mathcal{V}$ (D3) over a bounded duration t — the adversary's **coercion-time budget**, the maximum wall-clock time for which it can sustain coercion, calibrated against the incident durations in the registry [2]. Cooperation is genuine throughout by D3; the one exception, feigned cooperation, is confined to the residual window (§4.4).

2.2 The feasibility line

Asymptotic “hard” and “easy” are useless against a fixed, resourced attacker; these stipulations replace them.

Stipulation 1 (Feasibility line). *Let a cheap evaluation be one call to a non-memory-hard hash H , and a hard evaluation one call to an hours-scale Argon2-class function H^* whose cost is a large number of non-parallelizable steps — memory- and depth-hard,² so $\mathcal{A}$ ’s bounded speed advantage (D2) cannot amortize it [10].*

S1 (feasible). 2^{32} cheap evaluations are within $\mathcal{A}$ ’s budget.

S2 (infeasible). 2^{64} hard evaluations — written 64^* — are beyond it.

S3 (infeasible). 2^{96} cheap evaluations are beyond it.

The star denotes this memory- and depth-hardness and attaches to any bit-count — $32^, 64^*, 96^*, \dots$ — each meaning that many H^* evaluations, not just 64^* . Neither S1 nor S2 settles the star at 32 bits, so we state it: 32^* **is feasible** — memory-hardness alone does not lift a single 32-bit stage over the floor, which is why no orbit link commits fewer than 64 bits (§4).*

The floor. A work cost is *prohibitive* — equivalently, *meets the floor* — when it satisfies S2 or S3. The two are independent and we fix *no* ordering between them: for realistic hardness the per-evaluation * premium is well under 2^{32} , so the cheap bound binds, but * is user-tunable and an aggressive setting reverses that. Every “floor” claim names *this* threshold, not a fixed number, and holds whichever bound binds.

The star is kept rather than folded into a bit-count because memory- and depth-hardness has three consequences and a bit-count absorbs only the first: it inflates the brute-force cost, it imposes a *derivation time* per evaluation that the coercion window cannot outrun, and it rules out the Grover speed-up (§7). *Entropy* buys $32 \rightarrow 64$, but plain 2^{64} cheap work still sits at the border of a resourced attacker’s reach, so $64 \rightarrow 64^*$ must be bought by *memory-hardness* — which is why the gating function has to be Argon2-class. Plain cheap 2^{64} is the *softest number* throughout: named wherever invoked, never silently treated as infeasible.

2.3 The tacit-knowledge premise

Assumption 3 (Tacit knowledge, TKBA). *The secret is a set of fractal locations recognizable but not transmissible: $TKBA_1$, a point is non-transmissible without the fractal parameters that render it, so $\mathcal{V}$ cannot dictate it in the abstract; and $TKBA_2$, the deep fractal parameters are not a hand-overable artifact, since producing them requires walking a memory-hard chain, before which later locations do not exist. §4 instantiates the premise.*

² *Depth-hard* is meant more sharply than *sequential* alone: the cost is a non-parallelizable computation with a deliberately large number of steps — a long sequential *depth*, the parallelism-resistance axis of *depth-robust* memory-hard functions [9] — which $\mathcal{A}$ can neither parallelize away nor out-clock past (D2).

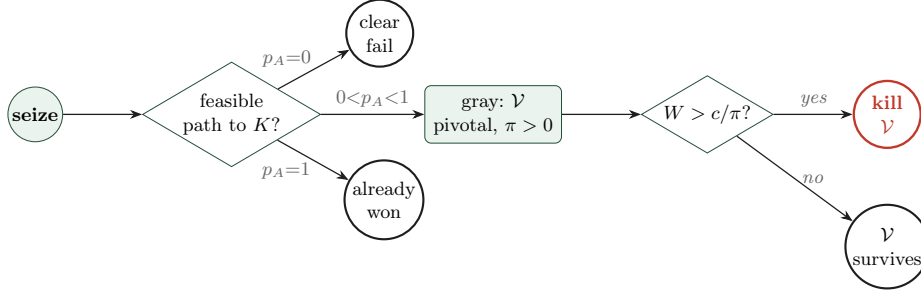


Fig. 1. The Deadly Race decision. At the *clear* endpoints ($p_A \in \{0, 1\}$) no race exists. In the *gray* interval the holder is *pivotal* ($\pi > 0$, Theorem 1), so once $W > c/\pi$ a rational $\mathcal{A}$ has a positive incentive to *assassinate* (Lemma 1). Clearing the Criterion means removing the gray branch entirely.

3 The Deadly-Race impossibility chain

“Nothing was your own except the few cubic centimetres inside your skull.”
— George Orwell, *Nineteen Eighty-Four*

The core negative result is an *escalating chain*: an *a fortiori* assumption on the attacker’s knowledge, its decision-theoretic consequence for a rational coercer, and the design bar the two force. Which links are *impossibilities* and which are *incentive* claims is stated throughout.

3.1 Setup

Fix the model of §2. Write K for the spend authority protecting an asset worth W to the holder $\mathcal{V}$, and c for the expected cost to $\mathcal{A}$ of killing $\mathcal{V}$. Bitcoin makes the prize *rivalrous* and *bearer*: the *first* party to broadcast a valid spend takes W ; one prize, no appeal.

Definition 1 (Seizable state, feasible path, racers). *The **seizable state** Σ is everything $\mathcal{A}$ obtains from the encounter: seized device state, plus whatever it extracts from $\mathcal{V}$ during the bounded coercion window. A **feasible path** from a state is an algorithm outputting K within the feasibility line (Stipulation 1). If $\mathcal{A}$ releases $\mathcal{V}$, two **racers** may pursue K — $\mathcal{A}$ from Σ , $\mathcal{V}$ from memory — and $p_A \in [0, 1]$ is the probability $\mathcal{A}$ spends first. The prize being rivalrous, $\mathcal{V}$ spending first to safety is $\mathcal{A}$ ’s only loss condition once it holds any feasible path.*

Definition 2 (Gray vs. clear outcomes). *An outcome is **gray** if afterward $0 < p_A < 1$: $\mathcal{A}$ has a feasible path yet $\mathcal{V}$, alive and free, can still win the race. The **clear** outcomes are the endpoints, $p_A = 1$ and $p_A = 0$ (Figure 1).*

The **priced** adversary $\text{WDY}[W, c]$ is a WDY adversary additionally endowed with (W, c) and modeled as a cold expected-utility maximizer over release, continued coercion and assassination; plain WDY carries no such payoff, so the

priced form is invoked only where a *quantitative* incentive is computed. Plain WDY worst-cases only the *payoff* coordinates ($W \rightarrow \infty, c \rightarrow 0$), leaving *capability* pinned at the feasibility line and the registry-calibrated coercion window: sending either to ∞ would collapse the floor or walk any time-gate to completion, a category error rather than a stronger guarantee (Appendix B).

3.2 The Non-Negative-Proof Principle

Assumption 4 (Non-Negative-Proof Principle, NNPP). *Kerckhoffs (Assumption 1) grants $\mathcal{A}$ the protocol — and that $\mathcal{V}$ may hold brain-memory or an undisclosed backup short-cutting K — but not how many such backups $\mathcal{V}$ keeps, or where. NNPP covers exactly that gap: what $\mathcal{A}$ cannot know is whether any undisclosed backup actually exists. The two do not conflict — one is the public mechanism, the other $\mathcal{V}$ ’s private state — so NNPP qualifies Kerckhoffs, and we resolve the gap conservatively: **$\mathcal{A}$ always assumes an undisclosed backup exists.** Something as small as an SD card, as inconspicuous as steganographic notes in a book, or as interior as memorized email credentials suffices to hold a short-cutting state.*

Both halves point the *same* way — toward killing — since each *raises* $\mathcal{A}$ ’s assessed payoff and hence its proclivity to kill; each is therefore *a fortiori*. In practice NNPP reduces to a plain observation: **exhaustive search is so ineffective that $\mathcal{A}$ always assumes another backup exists, whatever the terrorized $\mathcal{V}$ pleads.** It is rational, not paranoid — non-possession is a *negative existential* that cannot be certified on either reading of “prove,” the search reading failing because its last, irreducible site is the “few cubic centimetres” of the epigraph. An $\mathcal{A}$ that can read that space is a degenerate fixed point of the model: under an uncontested overlord there is no custody left to resist, so the only world falsifying NNPP is one in which the question does not arise (Remark 6).

Corollary 1 (Cooperation cannot dispel the incentive). *Let $\mathcal{V}$ hold b resumable copies and, under the coercion oracle, disclose some. Each disclosure confirms a copy existed, raising a lower bound $\hat{b} \leq b$; none yields the upper bound $\hat{b} \geq b$, which would need the covering search just ruled out. Disclosure is a monotone lower-bound estimator — “ $\mathcal{V}$ has revealed everything” is exactly the direction NNPP forbids. Hence no degree of cooperation drives $\mathcal{A}$ ’s posterior on “ ≥ 1 undisclosed copy” to zero: in any design leaving a released $\mathcal{V}$ a resumable path, the incentive of §3.3 cannot be discharged by surrender, only by the victim’s elimination.*

3.3 The Deadly-Race Lemma

Lemma 1 (Deadly-Race Lemma). *Let a custody scheme admit a gray coercion outcome (Definition 2): on release, $\mathcal{A}$ holds a feasible path to K ($p_A > 0$) while $\mathcal{V}$ retains one too ($p_A < 1$). Then $\text{WDY}[W, c]$ has a **strictly positive incentive to eliminate** $\mathcal{V}$ (and any co-racing heirs) whenever $(1 - p_A)W > c$.*

Proof. With $\mathcal{V}$ alive and free, $\mathcal{A}$'s expected take from the race is $p_A W$. By NNPP (Assumption 4) $\mathcal{A}$ cannot obtain or demand a verified assurance that $\mathcal{V}$ will not race — no such proof exists — so its *only* instrument for pushing the win probability toward 1 is to remove the competing racer. Eliminating $\mathcal{V}$ drives that probability to ≈ 1 and the expected take to $\approx W$. The marginal gain is $(1 - p_A)W$, and a rational $\mathcal{A}$ acts on it exactly when it exceeds c . In any gray outcome $p_A < 1$, so the gain is positive; for assets large relative to c the inequality holds.

The result is *decision-theoretic, not cryptographic*: the danger is created by the *shape* of the recovery path, not its hardness. And it *peaks where the marketing promises safety* — a “graceful” fallback (a slow-but-feasible recovery on a seized device, a time-locked rescue, a delayed vault) holds p_A well below 1 while leaving W fully at stake, i.e. it *maximizes* $(1 - p_A)W$. Graceful degradation, a virtue everywhere else, is here a liability — hence the title.

Remark 1 (Release is the trigger). The incentive is a property of the released gray state: it is $\mathcal{V}$ -as-free-racer whom $\mathcal{A}$ is paid to remove. A scheme that simply hands $\mathcal{A}$ the funds outright kills no one; the gray fallback is *worse than* that clean loss. And by Corollary 1 no amount of surrender removes the incentive, so only the victim's elimination closes it.

3.4 The No-Gray-Area Criterion

Criterion 1 (No-Gray-Area Criterion). *Under a fixed threat model, a coercion-safe scheme must admit **no gray outcome**: every encounter must resolve to a clear endpoint — $p_A = 0$ (the attack plainly fails: Σ feasibly reaches nothing and $\mathcal{V}$ holds nothing transmissible left to surrender) or $p_A = 1$ (it plainly succeeds: $\mathcal{A}$ obtains K within the encounter, so no post-release race exists). The forbidden interval $0 < p_A < 1$ is exactly where Lemma 1's incentive lives.*

The Criterion is a **bar, not a construction**, and its two endpoints are not equally desirable — $p_A = 1$ costs $\mathcal{V}$ the funds — but both are *survivable*: coercion resistance is properly a bar on **the holder's safety** first and the funds only after. Nor need it stay a stipulation: cast the encounter as a one-move game and no-gray *is* the safety condition.

Fix a scheme Σ and a priced adversary. In the one-move game **Coerce** $_{\Sigma}$ (Definition 3, Appendix A) $\mathcal{A}$ mounts an encounter (D3–D7), reaching a state with a set R of *racers* — the released $\mathcal{V}$ and any remote delegate — then chooses **spare** or **kill**, the latter removing $\mathcal{V}$ from R at cost c , and the race resolves with payoff $p_A(S_a)W - c \mathbf{1}[a = \text{kill}]$. Here $p_A(S)$ is $\mathcal{A}$'s probability of prevailing against racer set S , monotone in S , and by reachability $\mathcal{A}$ can eliminate only racers it can *reach*, never a remote delegate. Σ is **coercion-safe** if no priced adversary, at any reachable state, strictly prefers kill.

Theorem 1 (No-Gray-Area, as a theorem). *Write $\pi := p_A(R \setminus \{\mathcal{V}\}) - p_A(R) \geq 0$ for $\mathcal{V}$'s pivotality. Σ is coercion-safe against every WDY $[W, c]$ if and only if $\pi = 0$ at every reachable state.*

Proof. The kill increment is $\Delta U = \pi W - c$. ($\Leftarrow$) If $\pi = 0$ everywhere then $\Delta U = -c < 0$ and no priced adversary prefers kill. ($\Rightarrow$, contrapositive) If $\pi > 0$ at some reachable state then $\Delta U > 0$ for any (W, c) with $W > c/\pi$; that adversary exists and strictly prefers kill.

An experiment-and-advantage restatement, with the clauses D0–D8 as oracles and *reaching a gray state* as the winning event, is in Appendix A. Pivotality has exactly the two clearance routes of Corollary 2: under R1 ($R = \{\mathcal{V}\}$) removing $\mathcal{V}$ leaves no racer, so $\pi = \mathbf{1}[p_A > 0] - p_A$, vanishing *iff* $p_A \in \{0, 1\}$ — Definition 2’s no-gray condition as a special case; under R2 killing $\mathcal{V}$ leaves the delegate D , so $\pi = 0$ *iff* D dominates $\mathcal{V}$ as a racer, and the state may be *gray* yet safe because the pivotal racer is unreachable. That is contingent — a co-located or merely *slower* D restores $\pi > 0$, which is the BitVault case (§5). What the single-shot model leaves open is the *utility* behind ΔU (§7), not the equivalence.

Grading protocols. Coercion-safety grades a protocol by how much of the adversary it withstands, and §5 uses that vocabulary: **naively** coercion-resistant if it resists only a *sub-Kerckhoffs* attacker, so that denial or decoy works just while the attacker does not know $\mathcal{V}$ ’s setup; **weakly** if a full WDY encounter does not by itself yield K but *may* leave a deadly-race state ($\pi > 0$), the secret withheld *during* coercion and the holder still unsafe *after* it; and **strongly** if it yields neither, which is exactly coercion-safety. Cutting across the three, a protocol is **time-gating** if reaching K requires completing a gate of cost G set against the coercion budget t — a *mechanism*, not a strength (Definitions 9–12, Appendix B).

3.5 Reachability — the two clearance routes

Corollary 2 (Reachability). *Lemma 1’s incentive is realizable only against a racer $\mathcal{A}$ can reach. A scheme can therefore satisfy Criterion 1 in exactly two ways:*

- (R1) remove the race — *arrange that no feasible seizure path exists, forcing $p_A = 0$ (custody can stay individual); or*
- (R2) remove $\mathcal{A}$ ’s reach over the winner — *make the party who will win the race a remote delegate beyond $\mathcal{A}$ ’s coercive reach (custody becomes shared, at the cost of a trusted party).*

These are the classification axis of §5 — *leave no race to run* vs. *put the racer out of reach*. NNPP does *not* weaken the latter, as one might expect from $\mathcal{V}$ ’s inability to prove no local coercible copy of the delegate’s capability exists: that route’s holder-safety is exactly $\pi = 0$, and a hidden backup changes nothing, because **assassination removes a racer but never extracts a secret**. What survives is a difference about *funds*, and a plainer objection — an access-dominating delegate *is not self-custody*, its clearance resting on D refusing a **legitimate owner pleading for their life** (D8), the same refusal that defeats $\mathcal{A}$ ’s coerced plea (Remark 7).

3.6 Geographic, delegated, or tacit — what forces TKBA

Ask *how* a time-gating protocol can attain $\pi = 0$ against a single WDY encounter, confined to one *site* (D4). There are exactly three mechanisms, and each spends a different one of the *Four Properties*:

- (1) knowledge-based authentication — the secret in memory alone, i.e. *non-geographic*;
- (2) individual custody;
- (3) non-obscurity — which *is* the Kerckhoffs premise (D0);
- (4) coercion resistance.

A delegate is **access-dominating** (Definition 13) if its latency to reach and act on K never exceeds $\mathcal{V}$'s, so removing $\mathcal{V}$ leaves the race unchanged — R2, at the cost of (2). A protocol is **geographic** (Definition 14) if K 's gating is partitioned across sites, so a single-site encounter has no feasible path — custody may stay individual, but K now requires locations, the negation of (1). It is **tacitly gated** (Definition 15) if it gates K behind a *non-transmissible* in-head input (Assumption 3), which a single encounter can neither extract nor reconstruct offline: $\pi = 0$ at a single site, with the secret in memory only, keeping both (1) and (2). The geographic route clears the bar honestly but buys safety in *physical* coin — sites to burgle, a victim to be moved between them — so security rests on the physical defenses: Bitcoin reduced to a *glorified gold* guarded in vaults. The tacit route escapes that regress. Appendix B develops the geographic attack surface, the *literal* and *cyber-physical* races, checkable dominance, and why a perceptual oracle is worth its price only on the tacit route (Remark 13).

Theorem 2 (DRL-safety trichotomy). *A time-gating custody protocol is DRL-safe against a WDY adversary only if, at every reachable state, it is access-dominating-delegated (R2), geographic (R1), or tacitly gated (R1).*

The proof (Appendix B) turns on a two-blind-spots enumeration: a co-located WDY that seizes every device and coerces the present owner can be denied a piece of K 's gating only if that piece is *elsewhere* (geographic) or *in the unextractable mind* behind a time-gate exceeding the window (tacit); otherwise a racer at least as fast as $\mathcal{V}$ remains, which is the delegate. Kerckhoffs having thrown obscurity out, these are the only ways to restrict K 's availability at all. Of the three, the delegate forfeits (2) and the geographic route forfeits (1); **only the tacit leg keeps all four Properties**, and TGPO (§4) is that construction. The equivalence $\text{TKBA} \Leftrightarrow (1) \wedge (2) \wedge (3) \wedge (4)$ is Corollary 5 (Appendix B): (1) narrows the store to portable ones and (4) prunes those to the brain — every carried device is seized at the attack site, and a biometric is coerced — then forces its content to be non-transmissible. Unlike the trichotomy, it needs no exhaustiveness argument.

3.7 A fifth property — no hostage token

Devicelessness (1) yields a fifth property the Four leave implicit: immunity to indirect coercion *through a seizable token*. A *necessary* token stored outside the

holder’s head is a *hostage* — even where seizing K is infeasible, $\mathcal{A}$ can seize it, or threaten its destruction, and trade it for the formal, irrevocable transfer of another good. The property is orthogonal to R2 but *denied by construction* to any geographic protocol, which needs precisely such tokens; only the in-head secret has nothing to take hostage — and, under Kerckhoffs, its holder’s denial of a backup is *consistent with a perfectly ordinary setup* rather than a suspicious claim to be tortured out of (Remarks 14–17). The qualifier is load-bearing: leverage over a *person* is a different mechanism, and no custody design makes a relative unseizable (§5.5).

4 Construction — the Time-Gated Perceptual Oracle

The taxonomy leaves one route unexhibited. We close it with a construction clearing Criterion 1 by R1 — the *tacit* route that Corollary 5 shows is forced by holding all Four Properties at once: it admits *no feasible seizure path*, so custody stays individual. Call it the **Time-Gated Perceptual Oracle** (TGPO): *time-gated* for the memory-hard orbit, *perceptual* for the tacit fractal recognition, *oracle* for its evaluate-only-by-the-holder semantics. It is *one* way to clear the bar, not the paper’s contribution — §3 and §5 stand whether or not it does — so we give only what the floor theorem needs; Appendix C carries the orbit diagram, the proof and the residual analysis, and [11] the instantiation parameters.

4.1 What the construction is

The secret is a set of *tacit* fractal locations (Assumption 3): recognizable by the holder on a rendered board, but not dictatable, because a point cannot be transmitted without the fractal parameters that render it. Those parameters are themselves regenerated only by walking an *orbit* — a chain of N memory-hard (Argon2-class) stages seeded by a public, precomputation-resistant timechain salt [12] — so the deep boards do not exist until the derivation reaches them, and the raw state of each link is wiped in the instant between a cheap inner hash and the long memory-hard step, leaving seizable material live only momentarily. Each stage commits a whole Shamir polynomial [13] rather than a point, so no individual guess is ever checkable in isolation and $\mathcal{A}$ has no sub-threshold oracle. The master secrets are derived by a deliberately *cheap* hash at each stage, which keeps the terminal window instant and puts the resistance in entropy: a brute of the final stage from a bare board costs 2^{96} , infeasible by S3. Appendix C gives the orbit equations, the diagram, and the instantiation parameters are in [11].

4.2 The floor

Theorem 3 (The floor on the master secret). *Under Assumptions 2–3 and the standard build of (4)–(5), from every seizable state save a single residual window (§4.3) a target master secret K_N is prohibitive to reach — its cost meets the floor (Stipulation 1). A device seizure outside that window therefore*

yields nothing feasible: a clear attacker failure in the sense of Criterion 1, with no racer for $\mathcal{A}$ to reach.

The proof (Appendix C) is a case analysis over seizable states. Preimage resistance makes the orbit forward-only, so a seized board never yields its predecessor, and reaching the target from any earlier board requires guessing every intervening stage, each guess memory-hard-gated: pure device seizure leaves K unreachable, and a wrench of stage 0 or any mid-orbit seizure leaves K_N at 2^{96} or more. Only a live o_N escapes. Exercising the oracle on stages *below* the chain adds nothing to the floor (Remark 18); the reference implementation routes them through it anyway, to make the ceremony *observable* — showing a protocol-ignorant $\mathcal{A}$, on bits whose exposure costs nothing, that the rest of the derivation demands a faculty $\mathcal{V}$ has but cannot hand over (Remark 19). That moves $\mathcal{A}$'s *belief*, not the impossibility.

4.3 The tolerated residual seizure window

One sliver escapes, and it is first a *seizability* gap, not yet a deadly race: a wrench that lands with the final board o_N imminently or currently live and subdues $\mathcal{V}$ before the wipe catches the derivation essentially complete and $\mathcal{V}$ under control. There $\mathcal{A}$ *does* reach K and the stash is lost — we concede it plainly. What keeps it a *residual* is how narrow the coincidence is: a precise landing on the brief live- o_N interval, and a subdual before any wipe or reset.

4.4 The tolerated residual deadly-race window

Funds-loss is a *separate* axis from the deadly race, and the latter does not follow from the former. Killing buys $\mathcal{A}$ only a *shaving*: after the second-to-last point of stage N is surrendered, a cheap 2^{32} brute reaches K in ≈ 1 minute, versus waiting for a cooperating $\mathcal{V}$ to recollect it. The deciding quantity is therefore not $(1 - p_A)W$, non-negative by definition, but the *utility increment* $\Delta p_A W - c$ of assassinate-then-brute over keeping the victim cooperating. Appendix C shows it negative with overwhelming probability: the gain term is bounded by a minute the brute can be *pre-staged* around, the loss term is a strict subtraction because a victim killed before a wrong point is corrected is $\mathcal{A}$'s burned retry, and c is murder aggravated by frivolous motive *and* by the victim's defenselessness. This is an *incentive* argument against $\text{WDY}[W, c]$, not a theorem (§7), and the window is in any case a six-way coincidence — sophistication enough to drive the brute and the interface, a landing inside the o_N -live window, a subdual before reset, a judgment that immediate escape is critical, an ability to rule out a wrong surrendered point, and acceptance of a doubly-aggravated murder. Three design measures shrink it further: the anonymous time-lock cloaking *when* it occurs (Remark 20), canonical-island highlighting (Remark 21), and splitting a stash across several K_i (Remark 23); feigning is a bonus rather than load-bearing (Remark 22).

5 Classification of existing designs

Corollary 2 gives the classification axis: a design that clears Criterion 1 does so either by *removing the race* or by *removing $\mathcal{A}$'s reach over the racer*. This section places the shipping designs on it — stating each verdict and its reason, with the full analyses in Appendix D.

5.1 Time-lock-rescue vaults, and the reachable-racer failure

Rewind Bitcoin [14] holds funds in a time-lock vault: unlocking starts a multi-day countdown within which the owner *or a trusted delegate* may press “Rescue” and sweep to a cold panic address. Against *remote* theft this is sound. But the spend capability sits behind *only a delay*, keys on the seized device and no recall floor, so a wrench attacker holds a feasible path — unlock, wait out the countdown. That is a race, and Rewind’s answer is exactly R2: a remote delegate the attacker cannot reach presses Rescue. A legitimate clearance whose cost is that **custody is no longer individual**, and which holds only while the delegate is genuinely remote (Remark 24).

BitVault [15] is the same species. No authoritative specification is published; the account rests on the vendor’s pages, a self-attributed concept whitepaper [16] and correspondence, which do not agree in every particular. We note this to bound the claim, not as an allegation: we have no evidence that any material was withdrawn. Taking that whitepaper as valid, the spending paths carry timelocks — a primary needing a custodian co-signature, a one-year fallback on the user’s own two keys, a three-year custodian path — and both cases fail. Before the one-year date $\mathcal{A}$ coerces A and B and kills, the only party who would tell the gatekeeper to withhold being $\mathcal{V}$: removal is what buys settlement (Lemma 1). After it he coerces A and B and spends at once, ungatekept — no delay, no notification, nothing to cancel. §D.2 gives the sources and their limits. A caveat sharpens R2 itself: the winning racer must be not merely remote but *un-coercible* — an autonomous watchtower, not a human friend, since coercion routes through the victim to a person and a humane delegate pays (§5.5).

5.2 The industry recommendation set

The same structure appears in the *standard advice*. Threat-intelligence reporting states the objective in terms close to ours — custody should “remain resilient even when a human is under duress” — and recommends withdrawal delays, transaction-size thresholds, staged vault architecture, emergency freeze processes, and multisig with geographically distributed signers [6]. Criterion 1 does not reject this advice; it *partitions* it, and that partition is the framework’s practical content. Each control interposes a delay or a veto, hence defines a competing racer, so the only question is whether $\mathcal{A}$ can reach whoever holds the release. Where that authority rests with a remote, institutionally separated party, the control clears the bar by R2 at that route’s standing cost; where the delay is one the *holder* can release, or the freeze authority is a device or person on

the scene, it leaves a feasible-but-slower path — the gray outcome the Criterion forbids, and the configuration in which the control is worse than its absence. It is the reachability question, not the length of the delay, that separates a delay which buys rescue from one which buys $\mathcal{A}$ a reason to eliminate the other racer.

5.3 Self-destruct (wallet-erase) PINs

The designs above interpose a delay or a veto. A large and widely shipped class does something categorically different: it *erases*. Blockstream’s Jade documents such a PIN under both names its help center uses — *duress* and *wallet-erase* — as an alternate PIN that “automatically delete[s] the encrypted wallet information from Jade if entered,” “especially useful in the event of a physical threat, as you can provide this PIN to an attacker”; the device then reports only “Internal Error,” so the erasure reads as a malfunction [17,18]. Grant it the strength the delay architectures lack: it is *instantaneous and covert*, so $\mathcal{A}$ cannot intervene. The claim nonetheless decomposes into a real half and an obscurity bet — **(a)** the *erasure*, which happens whatever $\mathcal{A}$ knows, and **(b)** the *belief* that nothing further remains, manufactured by the “Internal Error” report. Only the erasure is Kerckhoffs-independent, and how much coercion resistance it carries alone is answered by the vendor’s own instructions: *none*. The same documentation warns that afterwards “there is no way to regain access to your funds except by restoring using your recovery phrase,” and directs the holder to “make sure you have a backup before proceeding” [18]. The feature *presupposes* a surviving backup: a holder who follows the documentation has one, and one who does not has destroyed the stash rather than defended it. What erasure removes is therefore not the secret but the *device path* to it — a path that cost $\mathcal{A}$ nothing to lose, since $\mathcal{V}$ is left alive, present and still able to restore, and one contributing nothing to the floor (Remark 18). The route that remains runs through $\mathcal{V}$ ’s memory of a seed phrase, or of where a backup is kept — **the erasure has deleted the path that required hurting no one and retained the path that does.**

That also makes it the worse of the two obscure bets it is sold beside, by an ignorance count (Remark 26): a decoy needs $\mathcal{A}$ ignorant only of decoys, and granted that it *terminates the encounter*, whereas a self-destruct needs him ignorant of the feature *and* of NNPP’s consequence — so even total success at the first leaves the encounter open, “this device is broken” inviting *where is your backup?* from a $\mathcal{V}$ still present and still knowing. It is the one obscure scheme with no branch on which it wins, and its own vendors document the regress: Coldcard’s firmware guidance, for the case where the duress wallet is disbelieved, advises escalating to the *brickme* PIN [19,20] — an irreversible erasure performed with $\mathcal{V}$ still in the room. The class lands in the **naïve** tier. Appendix D records what each of the three vendors shipping the pair actually *claims* for it (two decline the coercion claim on one of their two mechanisms), the two bluffs an informed $\mathcal{A}$ can play instead of entering any PIN, the companion decoy — sold as coercion protection and coached for believability (Remark 29) — and the *Obscurity Paradox*.

5.4 TGPO, and the honest comparison

TGPO clears by the *other* route: the secret is tacit and never on the device, so a seizure yields nothing feasible and there is no racer to reach. Read through the Four Properties the comparison is a covering-set statement — each shipping design trades at least one property TGPO keeps, and one trades a property and *still* fails.

Design	(1) Know.	(2) Indiv.	(3) Non-ob.	(4) Coerc.	What it trades
TGPO (this work)	✓	✓	✓	✓	holds all four (tacit; R1)
Rewind Bitcoin	×	×	✓	✓	individual custody: a remote delegate (R2)
BitVault	?	×	?	×	individual custody both ways — denial blocks the user, collusion spends past three years
Casa	✓	×	✓	✓	custody shared with remote co-signer
Self-destruct PIN	×	✓	×	×	non-obscurity: the erasure must read as malfunction (Colcard, Jade)

Row (1) is the covering point: TGPO is the *only* knowledge-based row, every competitor keeping a *transmissible* secret on a device. Casa’s collaborative multisig trades individual custody, a coercion-resistant delegate being remote *by definition* [21]. The axis tilts further to R1 for a blunter reason: the delegated rows (Rewind, Casa) are not self-custody at all (Remark 7).

5.5 Empirical grounding in the registry

The two stipulations leaning on the physical-attack registry [2] — the coercion-time budget and the murder-cost weighing of §4.4 — admit directional support. The registry records only *date*, *victim*, *location*, *headline*, so we code its 352 incident headlines (at commit 9a4a62a) by keyword; the coding is coarse and a reproducible script accompanies the paper.³ Four findings bear on the paper.

³ **registry-analysis/** gives the keyword patterns, per-year counts and caveats; Appendix D gives the method, the coding tables, the jurisdictional confounds and the case analyses.

The attack is frequent and worsening: recorded incidents climb from a handful a year before 2017 to a peak of 85 in 2025, and a separately sourced count — threat-intelligence reporting from law-enforcement disclosures and court documents — agrees, with 72 verified incidents in 2025, up 75% year on year [22], and 52 in the first half of 2026 against 39 a year earlier [6]. Reported *home invasions* there rose from 1 to 20 of 52: coercion is moving to where the holder lives, the setting the model assumes.

The terminal move is real: at least 5.1% (18/352) of headlines record a *killed* or *murdered* victim. A homicide is usually reported as a homicide rather than as a “Bitcoin attack,” so this is a *lower* bound — yet even so the outcome Lemma 1 predicts is attested in practice. That censoring compounds with a second: obscurity does not advertise itself, so no victim’s exposure to an obscurity-dependent design is on record either. The numerator is censored by how the terminal event is classified and the covariate by what vendors do not say, which is why the pattern cannot be read off the data and the bar has to be derived instead (Remark 31).

The mix is not universal, and the Lemma’s bite tracks it: by the ratio of long-hold to short-hold codings, France and the United States — the two largest national subsets, of near-identical size (61 and 59) — have *inverted* compositions, 7.0 against 0.7. Media inflation explains a *level* but not this *ordering*, and the difference matters for scope rather than validity: a hand-over-now robbery approximates the clean $p_A = 1$ endpoint, a multi-hour hold ending in deliberate release is exactly the gray outcome. The Criterion does its work where long holds dominate — and France alone accounts for a majority of the incidents recorded in 2026.

The attacker often does not need the holder at all: in 12 of the 352 incidents (3.4%, hand-coded and audited) the person seized was *not* a holder but a relative held to compel one, eleven in 2023 or later and nine in France. This is *not* the hostage token of §3.7 — the leverage is a person rather than a seizable object, so devicelessness does not reach it — but clause D8 failing in the field, and it bounds R2: the delegate a holder is most likely to appoint is precisely the party being seized, and a reachable delegate is not a second racer but a second victim.

Two fielded cases sharpen this, both analysed in Appendix D: the Balland abduction [23], whose rescue and recovery ran through exactly the state and centralized channels clause D8 assumes away; and the RATIRL torture [5], where a denial was *pre-discredited* because denial is the norm, so the rational continuation was *more* coercion. The second fixes the *sign*: obscurity does not merely fail to protect, it can convert a robbery into a torture.

6 Related work

Coercion resistance. The term is due to Juels, Catalano and Jakobsson [24], in electronic voting: a scheme is coercion-resistant if a voter *cannot prove* how they voted, so the coercer cannot verify compliance and the reason to coerce collapses. Our sense is complementary. There the protected act is a *choice* and safety comes from *unprovability*; here the protected object is a *bearer secret*, and

the danger is not unverifiable compliance but that a released holder remains a racer worth killing. We inherit the word, not the model.

Secrets recognized but not articulated. The tacit premise (Assumption 3) has direct antecedents. Bojinov, Sanchez, Reber, Boneh and Lincoln [25] train an authentication secret by *implicit learning* so that even a cooperating, coerced user cannot articulate it — the closest prior instantiation of our TKBA, with TGPO carrying the idea from an authentication token to a self-custody key behind a memory-hard time-gate. The broader phenomenon is old: keystroke dynamics treat typing rhythm as a biometric [26], and WWII traffic analysis identified wireless operators by the “fist” of their Morse keying [27].

Deniability versus unprovability. Deniable encryption [28] and the duress-PIN schemes it inspired let a coerced user *plausibly lie*. NNPP is the sharp negative counterpart: rather than proving a false positive (“this is the key”), the holder would need to prove a true negative (“no other key exists”), which cannot be done. Deniability offers a lie the coercer cannot disprove; we show the holder cannot furnish the assurance the coercer needs, which is why the coercer’s only remaining lever is the victim. Such schemes also rest on the adversary’s ignorance and so fail Kerckhoffs (§5.3).

Time-gates. The construction’s wall-clock gate is a time-lock puzzle in the sense of Rivest, Shamir and Wagner [29]; verifiable delay functions [30] are the modern publicly checkable form, and either may instantiate the outsourced time-lock the orbit optionally uses.

Kidnap economics and marginal deterrence. The Lemma’s shape is not new, and it would be wrong to present it as though it were. Selten’s kidnapping game [31] is the canonical formalization of the encounter we model, and the criminological literature on *marginal deterrence* — Stigler [32], formalized by Shavell [33] — has long carried the observation that an offender facing no incremental sanction has no reason to stop short of the worse act, with *witness elimination* as its standard illustration. At that level of abstraction the Deadly-Race Lemma is a witness-elimination argument in which what the victim retains is not testimony but a claim on a rivalrous prize. We claim no priority over either literature.

What is worth stating precisely is *where* our setting departs from Selten’s, because the departure is why our conclusion differs. Selten assumes the kidnapper’s threat to execute is credible *even though executing cannot improve his position* — correct in classic kidnap-for-ransom, where the prize is transferred by a third party and is not contestable once paid, so the hostage’s survival threatens nothing the kidnapper holds. Bitcoin self-custody removes exactly that property: the prize is rivalrous and bearer, so a design leaving $\mathcal{V}$ a feasible path leaves it *contestable after release*. In Selten’s terms our $(1 - p_A)$ is identically zero; the Deadly Race is his game with a non-degenerate post-release contest, and that single structural change turns execution from a threat instrument into an action with positive expected value. The contribution is therefore not the mechanism but its locus: marginal deterrence studies how a *legislator’s* penalty schedule

moves this margin, whereas a *custody designer* moves the same margin — and a widely shipped class of designs moves it the wrong way.

Two qualifications. Iqbal, Masson and Abbott [34] extend Selten by letting the capture probability rise when the hostage is executed, which is our c made explicit; our claim is that the gain side can exceed c for large W , not that c is small, which is why Lemma 1 is a threshold condition. And the deterrence literature in this vein — Lapan and Sandler on precommitment to non-negotiation [35] — addresses whether the encounter *occurs*, whereas our results concern the game once it has (§5).

7 Limitations and scope

The tacit premise is unproven — the softest assumption. Everything on the tacit route rests on a *perceptual oracle*: a stimulus a human can *recognize* but not *articulate*, which for TKBA must simultaneously admit a binary encoding, be feasible to memorize at ≥ 96 bits, and be infeasible to transmit within the coercion budget t (Definition 15). None of the three is proved here; their conjunction is an empirical, psychometric claim awaiting a field study. Perturbed Burning-Ship fractal locations are our instantiation, but until such a study **the validity of TKBA remains an assumption, not a result**, and it is the softest premise in the paper. Its imperfection is precisely the residual window (§4.3).

Living with it is affordable, and the objection has an empirical answer. Whether carrying ≥ 96 bits of tacit recall is too costly is a separate question from whether the oracle exists; retention is *assisted* [36], and the feat is historically ordinary. The objection’s stronger form — *users will never accept this much ceremony* — met a natural experiment in the July 2026 seed-entropy incident (§8): seeds made with 50-plus fair dice rolls were not at risk per the vendor’s advisory [37], and within days community guidance converged on that labor-intensive procedure. A second datum sits in the market for the rival method: holders pay for consulting that builds a decoy, and rehearse the story until it is believable (Remark 29). Both data are anecdotal and neither burden is TKBA’s, but together they locate the asymmetry — TKBA’s efficacy is conjectural, resting on a premise no study has established, whereas the decoy’s is condemned in theory by Kerckhoffs and NNPP and in the record by the registry (§5.5) — a record obscurity itself thins, which is why the condemnation has to rest on the theory (Remark 31). Effort tolerance is not a constant of the population; it reprices once a threat is legible, and it is already being spent — on the one of the two that theory and the record jointly condemn. Making the wrench-attack class legible *before* the corresponding incident is what a formal bar is for (Appendix F).

Quantum. A quantum search over an n -bit space needs $\sim 2^{n/2}$ *coherent* oracle evaluations [38], each here a deeply sequential Argon2d computation whose full *depth* the coherence must survive — a cost the resource-estimation literature finds decisive against far cheaper oracles [39]. A few-minutes H^* already places

Grover out of reach, so sub-256-bit standards are quantum-defensible *without* a 256-bit seed. The one bound Grover dents — the cheap 2^{96} brute, halved to $\sim 2^{48}$ — is confined to the residual window and buys nothing the classical 2^{32} brute there does not (Appendix F).

Stipulations aligned with standard practice. Two quantities are stipulated, not derived. The floor rests on the feasibility line (Stipulation 1) — a standard *key-size* stipulation; we claim no reduction to a named hard problem, and none is needed for key sizing. The coercion-time budget t is read off the registry [2], with the monotonicity that a kidnapping’s feasibility *falls* with its duration, so a *long-hold* adversary sustaining coercion past the gate is out of scope: against unbounded patience no in-head time-gate suffices. Most recorded attacks are short hand-over-now events (§5.5).

Single-shot utility, and why it is the conservative case. The kill decision is modeled once, risk-neutrally — *worst-for-victim*, since a risk-averse $\mathcal{A}$ discounts the uncertain gain and kills *less* readily. *Repeated* encounters are absorbed by the re-key pathway: a victim who survives with K uncompromised has every incentive to re-key to a fresh setup. *Inheritance* is beyond scope except for its design target — the protocol should render *assassinate-then-coerce-heir* non-viable, simplest remedy a dead-man switch with heirs also TGPO-protected.

Also out of scope. Adversarial-agent scenarios; device side-channels during active tacit recall; long-horizon cryptographic erosion; and mass on-chain correlation beyond standard taproot privacy. Finally, the framework fixes a *bar*; it does not claim TGPO is the unique construction clearing it, and the residual window (§4.4) is a bounded sliver argued incentive-safe rather than eliminated — an incentive argument against $\text{WDY}[W, c]$, not a theorem, and turning it into a quantitative bound remains open.

8 Ethics and responsible disclosure

This is offensive-flavored security research with direct life-safety stakes: we formally derive an *assassination* incentive. We therefore state plainly why publishing it is, on balance, protective.

The incentive is disclosed, not created. The Deadly Race is a property of a situation a coercer already faces; naming it hands nothing to an attacker who, by Kerckhoffs, is already assumed to know the protocol and the landscape. What publication changes is that *defenders* can now measure a design against the bar and reject the ones that manufacture the incentive. We describe an incentive and the failure modes of existing defenses — not a targeting method, and no operational how-to.

Obscurity is not a sound reason to withhold. Not publishing analyses of attack vectors and the failure modes of deployed defenses is itself a bet on obscurity, and a defense that works only while the adversary stays ignorant provides something worse than defenselessness: a *false sense of security*. Empirically, the registry [2] and the accompanying economic analysis [7] show what Kerckhoffs predicts — wrench attacks are frequent and rising, and the en-

trenched counsel (*keep a low profile, never disclose ownership, deny if asked*) has not stemmed them. Security knowledge diffuses to criminal actors regardless of academic reticence; withholding it handicaps only the defenders.

The quiet years compound — a supply-side precedent. The ecosystem has already run this experiment on itself, on the *generation* side of the key lifecycle. A March-2021 firmware defect left a fallback software PRNG silently live as the production entropy path of a major hardware wallet (Coldcard) [40,37]; pseudorandom output passes every eyeball test, so the weakness was invisible to inspection, and it stayed quiet for over five years — until a first wave of sweeps drained $\approx \$70\text{M}$ from $\approx 1,200$ addresses in 41 minutes on 30 July 2026 [41]. Two lessons transfer verbatim. The quiet years were not *obscurity working*: the loss was contracted the moment the defect shipped, and every silent month enrolled more holders into the eventual sweep. And there was no “quiet fix” even in principle — a warning channel wide enough to reach the holders is wide enough to reach the attacker. Wrench-resistance sits in the same epistemic position as entropy health: a holder might not realise from outside whether their custody design protects or endangers them, so collective silence about deadly-race dangers compounds exposure the same way, while it too reads as prudence.

Framing and restraint. We keep the framing on protecting at-risk holders, disclose our own bounded residual honestly rather than overclaim, and treat the designs we classify as legitimate clearances by the other route, not targets.

9 Conclusion

The Deadly Race turns “graceful degradation” from a virtue into a liability whenever the degraded state is a feasible-but-slower path to the secret: that state is a race, and the race pays an assassination incentive. The No-Gray-Area Criterion is the resulting bar, and the reachability corollary shows exactly two honest ways to clear it — time-lock-rescue vaults removing the attacker’s reach at the cost of a trusted third party, and the TGPO orbit removing the race entirely, keeping custody individual behind a floor that leaves K prohibitive to reach from every seizable state bar a bounded residual (Theorem 3).

What devicelessness buys. Keeping the whole setup in the head is also what lets the argument stay *a fortiori* — granting $\mathcal{A}$ every advantage and still finding no gray state — and keeps the game single-shot rather than a tangle of sub-games: the deviceless holder has a minimal window in which K is seizable at all and a *separate* minimal window of deadly-race exposure, no hostage device and so none of the bluffing maze of ransoming a seized token, no trusted third party, and no binding to sites of custody.

Open items. Four remain: a richer utility model behind the single-shot kill increment; a quantitative bound for the residual window (§4.4); an $\mathbf{H}^*$ calibration against hardware; and, softest of all, a field study establishing that a transmissible-resistant perceptual oracle exists (§7).

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Availability. The reference implementation, the design documentation and the registry-coding script are all public; URLs withheld for review.

A A game-based formulation

The body states coercion-safety in a *decision-theoretic* register: the coercion game **Coerce**_Σ (Definition 3) asks whether a *priced* adversary $\text{WDY}[W, c]$ ever strictly prefers to kill. Readers from the cryptographic tradition will expect the complementary *experiment-and-advantage* form. We give it here — and, mirroring the two axes the residual splits into (§4.3 for funds, §4.4 for the deadly race), we give *two* experiments over one shared WDY interaction: a **seizure game**, whose winning event is the familiar “the adversary reaches K ,” and a **deadly-race game**, whose winning event is the novel “the adversary reaches a *gray* state” — the one that manufactures the kill incentive (Lemma 1), and which an adversary can trigger even when it provably cannot learn K .

Two adversaries, and why the primed one is the worst case. The seizure game keeps the body’s adversary $\mathcal{A}$ (holder $\mathcal{V}$): it wants the *money*, seeks K , and would kill only *instrumentally*, where the priced calculus $(1 - p_A)W > c$ makes a kill raise its take (Definition 8). Its question is fund-safety — *does $\mathcal{A}$ reach K ?* The deadly-race game instead posits a *hypothetical* adversary $\mathcal{A}'$ (holder $\mathcal{V}'$, primed to keep the two games apart) that wants the deadly race *for its own sake*: it plays to reach a gray state. This is deliberately the worst case for the kill question. Reaching a gray state is *necessary* for any kill to pay (Lemma 1); so if even the DR-hunting $\mathcal{A}'$ cannot reach one — the coercion-resistance result below, $\Pr[\text{win}] = 0$ — then the money-seeking $\mathcal{A}$, who would kill only instrumentally, never lands in a state where killing $\mathcal{V}$ pays. *$\mathcal{A}'$ trying and failing to manufacture a deadly race is exactly what certifies that $\mathcal{A}$ will not run one.* Throughout, p_A and π stay unprimed: they are the body’s state functionals (Definitions 1, 2), read off the resulting position, and each game is a different *condition* on them.

A.1 The one-move coercion game

§3.4 states this game in prose; here it is in full.

Definition 3 (Coercion game and coercion-safety). *Fix a scheme Σ and a priced adversary $\text{WDY}[W, c]$. In Coerce_Σ : (1) $\mathcal{A}$ mounts an encounter ($D3$ – $D7$), reaching a state with a set R of racers — the released $\mathcal{V}$ and any remote delegate — and $p_A(S)$ is its probability of prevailing against racer set S (monotone: fewer racers, weakly higher p_A), so Definition 1’s scalar is $p_A(R)$. (2) $\mathcal{A}$ picks $a \in \{\text{spare}, \text{kill}\}$, where *kill* removes $\mathcal{V}$ from R — by reachability it can eliminate only racers it can reach, never a remote delegate. (3) The race resolves, with payoff*

$$U_{\mathcal{A}}(a) = p_A(S_a) \cdot W - c \cdot \mathbf{1}[a = \text{kill}], \quad S_{\text{spare}} = R, \quad S_{\text{kill}} = R \setminus \{\mathcal{V}\}.$$

Σ is **coercion-safe** if no $\text{WDY}[W, c]$, at any reachable state, strictly prefers *kill*.

A.2 The shared interaction

Definition 4 (The WDY interaction). *Fix a custody protocol Π , a WDY adversary $\mathcal{X}$ (Assumption 2), and the honest holder $\mathcal{Y}$.*

1. **Setup.** *The challenger runs Π ’s honest setup for $\mathcal{Y}$, producing the master secret K , the co-located device and explicit-secret state D (vault and time-lock blobs, commitment scripts, review schedules), and the tacit state T (the holder’s recognition capability). Public parameters pp , including the salt σ , are given to $\mathcal{X}$ (clause $D0$).*
2. **Interaction.** *Within a single attack site, $\mathcal{X}$ queries, in any order:*
 - **Seize** — *returns the co-located device and explicit-secret state D (clauses $D4, D5$).*
 - **Coerce(t)** — *for a coercion budget t , returns every value $\mathcal{Y}$ is able to transmit under full cooperation (clause $D3$); the tacit state T is available only as an evaluate-only oracle the challenger runs on $\mathcal{Y}$ ’s behalf, never as a transmissible value (clauses $D6, D7$).*

- **Compute** — any computation on the feasibility line (clause D1): at most 2^{64} cheap and strictly fewer than 2^{64} memory-hard evaluations.

Any remote delegate is not coercible through $\mathcal{Y}$ (clause D8).

3. **Resulting position.** Let R be the resulting set of racers, $p_A = p_A(R)$ the probability $\mathcal{X}$ then reaches K (Definition 3), and $\pi = p_A(R \setminus \{\mathcal{Y}\}) - p_A(R)$ the holder’s pivotality. Each game below is a predicate on this (p_A, π) .

A.3 The seizure game — fund-safety

Definition 5 (Seizure experiment; seizure-resistance). The *seizure experiment* $\text{Sz-Exp}_\Pi(\mathcal{A})$ runs the interaction (Definition 4) with adversary $\mathcal{X} = \mathcal{A}$ and holder $\mathcal{Y} = \mathcal{V}$, and outputs 1 (“ $\mathcal{A}$ seizes”) iff $\mathcal{A}$ feasibly reaches K — equivalently $p_A > 0$ (Definition 1). Π is **seizure-resistant** if $\Pr[\text{Sz-Exp}_\Pi(\mathcal{A}) = 1] = 0$ for every WDY adversary $\mathcal{A}$.

This is the fund-safety / seizability axis — the game face of the floor.

Corollary 3 (TGPO is seizure-resistant up to the residual). The Time-Gated Perceptual Oracle (§4) is seizure-resistant except on the residual seizure window (§4.3).

Proof. By Theorem 3, from every seizable state save that window K_N meets the floor, so no feasible path to K exists and $p_A = 0$; hence Sz-Exp outputs 0. On the residual seizure window Sz-Exp may output 1 — the o_N -live coincidence where $\mathcal{A}$ does reach K ; the loss there is of *funds*, bounded as argued in §4.3.

A.4 The deadly-race game — deadly-race-safety

Definition 6 (Deadly-race experiment). The *deadly-race experiment* $\text{DR-Exp}_\Pi(\mathcal{A}')$ runs the interaction (Definition 4) with adversary $\mathcal{X} = \mathcal{A}'$ and holder $\mathcal{Y} = \mathcal{V}'$, and outputs 1 (“ $\mathcal{A}'$ wins”) iff $\pi > 0$; for the individual-custody case $R = \{\mathcal{V}'\}$ this is exactly a gray outcome, $0 < p_A < 1$ (Definition 2).

Definition 7 (Coercion resistance). Π is **coercion-resistant** if

$$\Pr[\text{DR-Exp}_\Pi(\mathcal{A}') = 1] = 0$$

for every WDY adversary $\mathcal{A}'$.

Remark 2 (Where the two games part — the clean loss). The two experiments read the same position (p_A, π) but score it differently, and they diverge at exactly one point. At $p_A = 0$ both output 0 (no feasible path: the defender clears both). On the *gray* interval $0 < p_A < 1$ both output 1 ($\mathcal{A}$ holds a path and $\mathcal{V}'$ is pivotal: the defender fails both). They part only at *full compromise* $p_A = 1$: the seizure game outputs 1 (K is gone) while the deadly-race game outputs 0 ($\pi = 0$ — with the adversary already certain to win, removing the holder changes nothing, so no kill incentive arises). That single disagreement is the

whole content of “*a clean loss is deadly-race-safe*” (Remark 7): losing the funds outright is a seizure-game defeat but a deadly-race-game *non*-event. Conversely — the paper’s negative result restated — an adversary that provably *cannot* win the seizure game (cannot reach K ; e.g. one facing a graceful-recovery vault it cannot open within the custody window) can still win the *deadly-race* game by reaching a gray state. The deadly-race axis is therefore *not* implied by fund-safety; it is a separate axis, exactly as §4.3 and §4.4 are separate. Fund-safety the construction handles on its own terms — *on top of* removing the gray race, it minimizes the window in which the stash is seizable at all, via the wipe-before-memory-hard step of (4) and the floor (Theorem 3).

Remark 3 (The residual window: from 0 to ε). Definition 7 is the information-theoretic, no-gray-area form. The residual window (§4.4) replaces the exact 0 by a bounded $\varepsilon = \Pr[\text{Corr} \cap \text{Esc}]$: call Π ε -**coercion-resistant** if $\Pr[\text{DR-Exp}_\Pi(\mathcal{A}') = 1] \leq \varepsilon$ for all $\mathcal{A}'$. The claims below are the $\varepsilon = 0$ statements; the residual makes them ε -small — not zero — in the sliver argued incentive-safe there. In plain terms, this sliver is the moment the attack coincides with a *live* final orbit point o_N — currently or imminently materialized — and $\mathcal{A}'$ prevents the wipe (or reset) that would otherwise close it; only then is transmissible material briefly within reach, which is why the window is a bounded ε rather than a standing exposure.

A.5 Consistency with the main results

Theorem 4 (The experiment coincides with the Criterion). *Π is coercion-resistant (Definition 7) if and only if $\pi = 0$ at every reachable state — equivalently, iff Π satisfies the No-Gray-Area Criterion (Criterion 1), iff Π is coercion-safe in the decision-theoretic game (Definition 3).*

Proof. DR-Exp outputs 1 exactly on a reachable state with $\pi > 0$, so $\Pr[\text{win}] = 0$ iff no reachable state is pivotal. By Theorem 1, $\pi = 0$ everywhere is equivalent to coercion-safety and to Criterion 1. The two clearance routes of Corollary 2 — individual custody (R1) and a remote un-coercible racer (R2) — are exactly the two ways to force $\Pr[\text{win}] = 0$.

Corollary 4 (TGPO is coercion-resistant). *The Time-Gated Perceptual Oracle (§4) is coercion-resistant.*

Proof. The experiment outputs 1 only on a *gray* state ($\pi > 0$); both clear endpoints $p_A \in \{0, 1\}$ give $\pi = 0$ and hence output 0. By Theorem 3 every state reachable by **Seize**, **Coerce**, and **Compute** leaves K at the floor, so the reachable endpoint is the *clear failure* $p_A = 0$ — the opposite endpoint $p_A = 1$ would require feasibly reaching K , which the same theorem bars. Either way $\pi = 0$, so the experiment never outputs 1 and $\Pr[\text{win}] = 0$. The only place p_A rises above 0 (to 1 in the worst instant) is the residual window (§4.3), where the tacit premise momentarily fails; accounting for it yields ε -coercion resistance.

Remark 4 (Reading the classification as adversaries). Under Definition 7, the classification of §5 is a statement about which designs admit a winning $\mathcal{A}'$. Against a time-lock-rescue vault advertised as *individual* custody, the adversary that calls **Seize** and then waits out the delay is winning: the rescue racer is co-located and reachable (clause D4), so $0 < p_A < 1$ and $\pi > 0$. Routing rescue to a genuinely remote, un-coercible delegate removes the reachable racer and restores $\Pr[\text{win}] = 0$ — the honest R2 case, at the cost of individual custody.

B Supplement to the impossibility chain

This appendix carries the full statements that §3 summarizes: the priced-adversary family and its worst case (§B.1), the self-defeat argument behind NNPP (§B.2), the grading definitions the taxonomy of §5 uses (§B.3), the honest cost of the remote-delegate route (§B.4), the three DRL-safe mechanisms and the variants of the perceptual oracle (§B.5), the Four Properties characterization (§B.6), and the hostage-token property (§B.7). Nothing here is needed to follow the argument of §3; it is the detail a reader who wants to check that argument will ask for.

B.1 The priced adversary

Definition 8 (Priced and sharpened adversaries). *The **priced** WDY adversary, written $\text{WDY}[W, c]$, is a WDY adversary (Assumption 2) additionally endowed with the economic parameters (W, c) and modeled as a cold expected-utility maximizer: it chooses among release, continued coercion, and assassination so as to maximize its assessed expected payoff over the deadly-race probabilities (the win probability p_A , and the correctness/escape events introduced in §4.4). Plain WDY carries no such payoff: it is the adversary of the general results (NNPP, the No-Gray-Area Criterion, the floor), which must hold whatever (W, c) turn out to be. $\text{WDY}[W, c]$ is invoked only where a quantitative incentive is actually computed — the Deadly-Race Lemma’s condition (§3.3) and the residual-window analysis (§4.4).*

A third, optional parameter sharpens the feasibility line to a single threshold. $\text{WDY}[W, c, b]$ replaces the two stipulations of Stipulation 1 by one clear-cut bound b — total work below b feasible, at or above b infeasible — in cost-homogeneous units, with memory-hardness folded into the per-evaluation cost. This idealization is legitimate here because the protocol quantizes entropy in 32-bit chunks, so feasibility is a step function on the lattice $\{32, 64, 96, \dots\}$ and its hard counterparts $\{32^, 64^*, 96^*, \dots\}$: a threshold at that granularity loses nothing, whereas pricing a single brute attempt would be finer than any distinction the construction can express. Conservatively $b = 64^*$ — work below 64^* feasible, 64^* itself infeasible. A fourth optional parameter t budgets the **coercion time** (D3’s window): $\text{WDY}[W, c, b, t]$ fixes the maximum wall-clock time for which the adversary can sustain coercion. Like b , t is a capability bound pinned at its real, registry-calibrated value t_{reg} , not sent to ∞ — unbounded coercion would let $\mathcal{A}$ walk*

any time-gate to completion, the same category error as $b \rightarrow \infty$. The rungs are thus WDY (capabilities only), WDY[W, c] (+ payoff), WDY[W, c, b] (+ sharp feasibility), and WDY[W, c, b, t] (+ coercion-time budget).

Remark 5 (Plain WDY as a worst case). Plain WDY is exactly the worst case of the priced family over the *payoff* coordinates, taken at the model’s *fixed* capability:

$$\text{WDY} = \text{WDY}[W \rightarrow \infty, c \rightarrow 0, b = 64^*, t = t_{\text{reg}}].$$

The payoff coordinates are sent to their extreme — infinite stake, costless homicide: the maximally kill-prone attacker, which assassinates in *every* gray outcome (§3.3) — because the general results must hold whatever (W, c) are. The capability coordinates are **not** worst-cased: b stays pinned at the feasibility line 64^* and t at the registry window t_{reg} . Sending $b \rightarrow \infty$ would grant $\mathcal{A}$ a 64^* brute and collapse the floor (Theorem 3); sending $t \rightarrow \infty$ would let it walk any time-gate to completion and collapse the tacit route (§3.6). Either is a category error, not a stronger guarantee. The asymmetry has a tell at the residual window: there $p_A \rightarrow 1$ while the worst case wants $W \rightarrow \infty$, making $(1 - p_A)W$ an $\infty \cdot 0$ indeterminate form — which is precisely why that window is settled with *finite* (W, c) and the Δp_A increment (§4.4), not with the blunt limit.

B.2 Why even total search fails

Remark 6 (Why even total search fails — the self-defeat). The search reading fails not merely because the covering is large but because the capability that would complete it is *off the domain of the problem*. To certify $\neg \exists$ witness, $\mathcal{A}$ must reach the last, irreducible site — the “few cubic centimetres” of the epigraph: whether $\mathcal{V}$ has *forgotten*, or never committed, K to memory. An $\mathcal{A}$ that can read that space, and cover all physical hiding-space besides, is *supra-totalitarian* — it exceeds the literary supremum of surveillance power, which expressly conceded the skull — and it is a *degenerate fixed point* of the model: it violates the bounded-adversary premise so completely that the contested-property premise collapses with it, for under an uncontested overlord there is no custody to resist. Defending against it is a *doomsday bet*: the branch in which the adversary exists is the branch in which the stakes that define the game are gone, so a rational actor assigns it *zero decision-weight* — not because its probability is negligible, but because its utility is undefined there. NNPP is therefore safe as a *modeling boundary*: the only world that falsifies it is one in which the question does not arise.

To avoid a nightmarishly complex bluffing game between $\mathcal{A}$ and $\mathcal{V}$, and to avoid encouraging real users to bet their stashes — and their lives — on it (see Remark 29), we also do not consider $\mathcal{A}$ reading truthfulness off $\mathcal{V}$ ’s terrorized demeanor or tone. The simplification this forces is that $\mathcal{A}$ is *uniformly distrusting*: it always defaults to assuming that whatever NNPP posits *might* exist does in fact exist. The one alternative simplification — a *gullible* $\mathcal{A}$ that always believes

$\mathcal{V}$ — is trivially sub-NNPP and sub-Kerckhoffs, hence useless as a threat model. Erring toward distrust is therefore the only valid choice, and it is what makes the model *a fortiori*: a protocol that delivers seizure resistance or deadly-race resistance against a uniformly distrusting attacker delivers them against a real one too.

B.3 Grading protocols: the four definitions in full

Definition 9 (Naively coercion-resistant custody protocol). *A protocol is **naively coercion-resistant** if it may resist a sub-Kerckhoffs attacker — one with the physical capabilities **D1–D7** but lacking Kerckhoffs knowledge (**D0**, Assumption 2) — because denial, decoy, or obscurity may work when the attacker does not know $\mathcal{V}$ ’s setup. Both hedges are load-bearing: the guarantee is contingent on the attacker’s ignorance, not on any structural property, so it offers no guarantee against the full WDY adversary. Examples: plausible-denial and decoy wallets, or time-gated transactions cancelable through an obscure key — each collapses the moment **D0** is granted.*

Definition 10 (Weakly coercion-resistant / weakly WDY[W, c, b, t]-resistant). *A protocol is **weakly coercion-resistant** (equivalently, **weakly WDY[W, c, b, t]-resistant**) if a single encounter with the full WDY adversary (**D0–D8**) does not, by itself alone, yield the master secret K , but may leave a deadly-race state: a reachable state of positive pivotality $\pi > 0$ (Theorem 1) in which $\mathcal{A}$ holds a feasible-but-slower path and $\mathcal{V}$ remains a reachable racer. The secret is withheld during coercion; the holder is not safe: after coercion finishes $\mathcal{A}$ may enact deadly race and ultimately seize K nevertheless.*

Definition 11 (Strongly coercion-resistant / strongly WDY[W, c, b, t]-resistant). *A protocol is **strongly coercion-resistant** (equivalently, **strongly WDY[W, c, b, t]-resistant**) if a single encounter with the full WDY adversary (**D0–D8**) yields neither K nor a deadly-race state — every reachable state has $\pi = 0$. This is exactly coercion-safety (Definition 3, Theorem 1): the bar Criterion 1 sets, and the target the construction of §4 meets.*

Definition 12 (Time-gating custody protocol). *A custody protocol is **time-gating** if reaching its master secret K requires completing a time-gate — a computation or delay of wall-clock cost G (a memory-hard derivation, or a time-lock) — so that K is unavailable before G elapses; G is the design parameter, set against the adversary’s coercion-time budget t (**D3**). Time-gating is a mechanism, not a strength: a given time-gated protocol instantiates one of the three levels above according to the outcome it forces — **naive** if it relies on an obscure self-destruct (the holder destroying the device’s own copy) or a cancel key (safe only against a sub-Kerckhoffs attacker); **weak** if the gate leaves a feasible-but-slower path, keeping $\pi > 0$; **strong** if every reachable state has $\pi = 0$ (Theorem 1), no matter the mechanism. It is the running example of the taxonomy.*

B.4 The honest cost of the remote-delegate escape

Remark 7 (The honest cost of the remote-delegate escape). It is tempting to say NNPP (Assumption 4) also weakens R2, on the grounds that $\mathcal{V}$ cannot prove that no local, coercible copy of the delegate’s rescue capability exists. It does not. By Theorem 1 the holder-safety of R2 is exactly $\pi = 0$, which holds *iff* D access-dominates $\mathcal{V}$ (Definition 13); a hidden backup changes nothing, because **assassination removes a racer but never extracts a secret**. Were such a copy extractable, $\mathcal{A}$ would take it *during* the encounter — a clean success ($p_A = 1$, no post-release race), the survivable endpoint; were it not, D still dominates and killing $\mathcal{V}$ is pointless. Neither branch is a deadly race. NNPP bites only in the *original* race (Lemma 1), where $\mathcal{A}$ *holds* a feasible-but-slower path from the seized device and cannot be reassured that $\mathcal{V}$ will not race; it does not reach the delegated case.

One honest residual difference survives, and it is about *funds*, not the holder. The delegate’s capability is *transmissible*, so R2 keeps a clean-loss exposure that R1 lacks: if $\mathcal{V}$ *did* retain a coercible copy, coercion extracts it and the funds are lost cleanly ($p_A = 1$). R1’s secret is *non-transmissible* (D6–D7, Assumption 3), so full coercion extracts nothing.

The decisive objection, though, is an elephant in the room: an access-dominating delegate *is not self-custody*. Its entire clearance rests on assuming D will **refuse a legitimate owner pleading for their life** (D8) — the very refusal that defeats $\mathcal{A}$ ’s coerced plea also defeats $\mathcal{V}$ ’s genuine one. Whether D ’s state is *sufficient* for K or merely *necessary*, shared custody reinstates exactly the trusted counterparty the design set out to remove. *Not your keys, not your coins*: that, more than any impossibility, is the standing case for the individual-custody route (R1).

B.5 The three DRL-safe mechanisms, and the perceptual oracle’s variants

Definition 13 (Access-dominating delegate). *For a party P let $\ell_P(\tau)$ be its latency to reach and act on K (spend, or rescue/cancel) from the state at time τ . A delegate D is **access-dominating** if $\ell_D(\tau) \leq \ell_V(\tau)$ at every τ — K is, at all times, at least as available to D as to the owner. Then D is never slower than $\mathcal{V}$, so removing $\mathcal{V}$ leaves the race unchanged, $p_A(R \setminus \{\mathcal{V}\}) = p_A(R)$, giving $\pi = 0$ (route R2). Operationally the owner periodically syncs D ’s state, holding $\ell_D \leq \ell_V$.*

Remark 8 (Dominance can be made checkable). The sync is best run during *peacetime* (the quiescent interval between encounters), and it can be made *verifiable*: two time-separated digests of D ’s state reveal not only that D leads but its rate v_D . Knowing its own rate v_A and the two remaining distances to K — Δ_{S_A} for $\mathcal{A}$, Δ_{S_D} for D — $\mathcal{A}$ can settle the race in advance: it can win only if $\Delta_{S_A}/v_A < \Delta_{S_D}/v_D$. A delegate running near the frequency ceiling (D2) against a slower $\mathcal{A}$ makes racing plainly futile — turning D ’s dominance from $\mathcal{A}$ ’s *credence* into something $\mathcal{A}$ can *check*. Whether D ’s held state is *sufficient* for K or

merely *necessary* (so that defeating $\mathcal{A}$ needs only that D can *block*, not spend) does not change the argument.

Definition 14 (Geographic protocol). *A protocol is **geographic** if K 's gating is partitioned across ≥ 1 sites, so that reaching K requires the co-located seizure and coercion of co-located principals (D3–D7) to be deployed at every such site. A single WDY encounter is confined to one site (D4); the moment K 's gating requires a site the encounter does not occupy, it has no feasible path, $p_A(\emptyset) = 0$, so $\pi = 0$ (route R1). Custody may stay individual (the other sites are the owner's own devices), but K now requires geographic locations — which is exactly the negation of Property 1 (knowledge-based $\equiv$ non-geographic).*

Remark 9 (The geographic route is legitimate but inherently physical). Definition 14 clears the bar honestly; but it does so by moving the burden onto *physical* distance and guarding, and an honest account should name the avenues it leaves $\mathcal{A}$ rather than gloss over them. Against a protocol whose K needs ≥ 1 site of custody, $\mathcal{A}$ may — singly or in combination:

1. **catch $\mathcal{V}$ at a site:** then the encounter's single site (D4) is a gating site, and only the *other* sites stay out of reach — which is why the definition needs a site the encounter does not occupy, and why a lone site is the weakest case;
2. **burgle the sites without $\mathcal{V}$,** taking the tokens directly — possibly after enacting a deadly race, since removing $\mathcal{V}$ (Lemma 1) buys uncontested time to travel and break in;
3. **burgle one gating site and keep what he takes:** a token K needs but $\mathcal{A}$ cannot use alone is leverage rather than a route — a *hostage token* (§3.7) held against $\mathcal{V}$ for the rest;
4. **abduct $\mathcal{V}$ and move them from site to site,** seizing each in turn — a chain of single-site encounters, bounded only by travel time and the coercion-time budget t (D3);
5. **coerce $\mathcal{V}$ into having allies retrieve and transmit** the remotely-kept tokens — excluded here by D8 (the remote party does not comply), but named for completeness.

None of these is defeated by *cryptography*: each is bounded, if at all, only by *physical* facts — how far apart the sites sit, how well they are guarded, how much travel the window t forbids. A geographic protocol is thus a *legitimate but inherently physical* clearance, and one that *scales in cost*: more sites, farther apart and better defended, cost more to hold and to guard.

Remark 10 (Literal Race). A geographic protocol (Definition 14) can turn the deadly race into a *literal* one: once the wrench attack ceases, whichever party — $\mathcal{V}$ or $\mathcal{A}$ — first roams to and reaps enough of the remote custody sites takes the prize W . The race is now run over physical ground rather than computation. This exposes a second, purely *physical* lever: depending on how well those remote sites are defended, $\mathcal{A}$ may assess its probability of successfully seizing the remote secrets — hence its pivotality π — as so low that $\pi W \leq c$, i.e. $W \leq c/\pi$; by Theorem 1 the (literal) deadly race is then *not worth enacting*.

Remark 11 (The geographic escape reduces to physical security). But note what that concedes: security now rests on the question “*how effective are the physical defenses against the assumed physical attack capability?*” — Bitcoin reduced to a *glorified gold* guarded in vaults (cf. the accompanying economic analysis [7]). The tacit route (R1, Definition 15) is precisely what escapes this regress: it reduces the physical-security problem to an *actual cryptographic* one, making banking-grade physically-secure self-custody available at the cost of running a piece of free open source software.

Remark 12 (Cyber-physical race). A protocol that admits *two or more* feasible paths to K — one running through the collection of physically distributed tokens and another that does not (say an undisclosed, locally hidden shortcut, or a state seized *during* the wrench attack) — yields a *cyber-physical* race: one racer works to complete a KDF (or time-lock) computation while the other roams the custody sites. The temporal and spatial clearance routes are then run in parallel by different parties, and p_A is fixed by whichever path is faster — a mixed regime that the pure-temporal (tacit) and pure-spatial (geographic) analyses each capture only half of.

Definition 15 (Tacit gating (TKBA)). *A protocol is **tacitly gated** if it attains coercion resistance by gating K behind a non-transmissible in-head input (Assumption 3): the deep perceptual stages are recognizable but not dictatable. A single encounter can then neither extract the input (non-transmissible, and the coercion-time budget t of $D3$ is too short to walk the derivation) nor reconstruct it offline (the seized state meets the floor), so $p_A(\emptyset) = 0$ and $\pi = 0$ (route R1) — now at a single site, with the secret in memory only. This keeps Properties 1 and 2. In TGPO the gate is temporal — a memory-hard chain with $G > t$ (§4).*

Remark 13 (The perceptual oracle is worth its price only on the tacit route). The perceptual gate need not be *temporal*. It could sit at a physically protected custody site (a *geography-gated* oracle), complete only with a remote delegate (a *delegate-gated* one), or be gated on knowledge of the protocol alone (an *obscure* one). None of the three is worth building, and the reason is the same in each case.

The oracle is expensive: it costs the holder the standing memorization burden §7 concedes, and that price buys exactly one thing — the *conjunction* of the Four Properties (Corollary 5). A variant that already relaxes one of the four has spent the price on the benefit it has just given up. A holder who can relax Property 1 should take a plain geographic protocol, which yields the other three at no oracle cost; one who can relax Property 2 should take a plain access-dominating delegate, on the same terms. In both cases the oracle is *dominated*, not merely unattractive.

Obscurity fails twice over. By the argument above, an $\mathcal{A}$ whose ignorance may be assumed makes the oracle redundant — simpler obscure schemes (denial, decoy, a self-destruct) already suffice. And if that ignorance may *not* be assumed — i.e. under Kerckhoffs (D0) — then the only thing gating the perceptual parameters is knowledge of their existence, so they are *available* at the

attack state; by D6–D7 they and their points are seized. That variant is not merely dominated but ineffective.

What survives is the temporal instantiation. The tacit route is the one place where the oracle’s price buys something no cheaper design delivers, which is why §4 gates it on a memory-hard chain and nothing else. The geographic and delegated *routes* remain legitimate clearances (Theorem 2); it is only a perceptual oracle riding on them that is pointless.

B.6 The Four Properties characterization

§3.6 states the trichotomy and its consequence for the Four Properties. The proof sketch and the characterization are these.

Proof (Proof sketch). By Theorem 1 DRL-safety is $\pi = 0$, i.e. $p_A(R \setminus \{\mathcal{V}\}) = p_A(R)$. Either $R \setminus \{\mathcal{V}\}$ retains a racer at least as fast as $\mathcal{V}$ — an access-dominating delegate (Definition 13) — or it does not, whence (dropping the degenerate clear-success $p_A(R) = 1$) $p_A(\emptyset) = 0$: a single encounter has *no* feasible path, so a piece of K ’s gating is withheld from an adversary that seizes every co-located device and coerces the present owner. A co-located WDY has only two blind spots — *elsewhere* (another site: geographic) and *the unextractable mind* (non-transmissible, behind a time-gate exceeding the window: tacit). No third withholding exists.

Corollary 5 (TKBA characterizes the Four Properties). *Necessity.* *Property 1 demands a store of maximal availability — one that travels with the owner, not a fixed location. Several qualify: the brain, but equally a body-implemented chip, a biometric token, or a USB stick carried everywhere. Property 4 prunes them: every carried device is co-located at the attack site, hence seized (D4), and a biometric is coerced — only the brain, the one store that is not a device, survives. The brain still holds both coercible (explicit, memorizable) and non-coercible content, so the same Property 4, now against the coercion oracle (D5), forces the secret to be the latter: non-transmissible, i.e. tacit. Thus (1) $\wedge$ (3) $\wedge$ (4) yield a tacit brain secret, and (2) that it is the owner’s alone — TKBA. **Sufficiency** is the reverse check — TKBA’s input is in memory (1), needs no second party (2), resists a protocol-aware WDY by non-transmissibility rather than a hidden trick (3), and forces $p_A(\emptyset) = 0$ up to the tolerated residual (4). Hence*

$$TKBA \iff (1) \wedge (2) \wedge (3) \wedge (4).$$

Of the three DRL-safe routes (Theorem 2), the delegate forfeits (2) and the geographic route forfeits (1); only the tacit leg keeps all four, and TGPO (§4) is that construction.

This characterization needs *no* exhaustiveness argument: Property 1 narrows the store to portable ones, and Property 4 pins it to the brain and its content to non-transmissible. The *general* trichotomy does rest on a two-blind-spots enumeration; the Four-Properties equivalence stands independently of that step.

B.7 The hostage-token property

Remark 14 (Indirect wrench via a hostage token). Even a device-dependent setup whose K is deemed unwrenchable can present a surface for an *indirect* wrench attack through its most wrench-vulnerable device-stored token. $\mathcal{A}$ seizes that token and holds it hostage, to be traded for the formal transfer of another of $\mathcal{V}$'s assets — possibly one *less* liquid and *more* traceable, and so otherwise easier to rescue: a car, jewellery, real estate. For instance, take a setup in which $\mathcal{V}$ carries one necessary token at all times and keeps the other at a site beyond $\mathcal{A}$'s wrench reach. Caught *away* from that site, $\mathcal{V}$ can be relieved of the carried token and threatened with its destruction unless they *legally* — hence irrevocably — transfer some otherwise-rescuable asset. And a site of custody, not personally guarded by its owner, is itself a stationary (and possibly unguarded) surface for token seizure.

Remark 15 (Weaponizing the hostage to probe for backups). The same lever can be turned to *detect* what NNPP (Assumption 4) holds $\mathcal{V}$ cannot disprove. By how dearly $\mathcal{V}$ ransoms a hostage token, $\mathcal{A}$ reads off information about backups: a high rescue price paid suggests no backup to that token exists. This trades NNPP's clean simplification for a game-theoretic can of worms — either party may bluff — and lies beyond the scope of this work.

Remark 16 (Deviceless TKBA avoids the can of worms). Deviceless, classic TKBA has no external token to take hostage, and so escapes the whole tangle: with the secret in the head alone, there is nothing to seize, ransom, or read a backup off.

Remark 17 (Denial is inherently plausible). NNPP (Assumption 4) leaves $\mathcal{A}$ unable to disprove an undisclosed shortcut backup — but for the deviceless holder that cuts the other way. Under Kerckhoffs $\mathcal{A}$ knows the protocol, and so knows a TGPO setup reaches K from the head alone: $\mathcal{V}$ has no *need* of a shortcut device. A denial — “there is no backup” — is therefore *consistent with a perfectly ordinary setup*, not a suspicious claim to be tortured out of. Where a device-dependent holder might plausibly keep one, the deviceless holder plausibly keeps none: the very absence of need is what makes the denial more credible, and consequently, $\mathcal{A}$ more inclined to desist.

C Supplement to the TGPO construction

This appendix carries the material §4 summarizes: the orbit diagram, the proof of the floor theorem, the null result on shallow-stage oracle use and the adversary-education argument beside it, and the full residual-deadly-race analysis with the design measures that shrink the window.

C.1 The orbit

Let H be a cheap, entropy-preserving one-way hash, H^* a memory-hard (Argon2-class) hash of tunable cost, and σ a public, precomputation-resistant *timechain*

salt whose cross-user amortization is negligible [12]. An **orbit** of N deep stages (Figure 2, Appendix C) is

$$o_0 = H(\sigma), \quad (1)$$

$$u_i = H(o_i \parallel Sh_i), \quad 0 \leq i \leq N-1, \quad (2)$$

$$o_i = H^*(u_{i-1}), \quad 1 \leq i \leq N, \quad (3)$$

$$K_i = H(o_i \parallel Sh_i), \quad 0 \leq i \leq N, \quad (4)$$

with $\{o_{i-1}, Sh_{i-1}\}$ **wiped** in the instant between the cheap inner hash and the long memory-hard step, so the raw prior orbit point (o_{i-1}) and Shamir secret (Sh_{i-1}) exist only momentarily rather than across the whole memory-hard window — which is why the construction does not use $H^*(o_{i-1} \parallel Sh_{i-1})$ directly. (u_i and K_i share a formula but differ by *role*, kept apart in deployment by domain separation.)

Each stage carries not a point but a *whole Shamir polynomial* [13]: s_i points $p_{i,j}$, each a 32-bit leaf-area of a fractal $\theta_{i,j} = H(o_i \parallel j)$, any r_i of which reconstruct Sh_i of $r_i \cdot 32 \geq 64$ bits. The orbit absorbs the *full* polynomial, so the smallest object any link commits is a whole stage and no individual point is ever checkable in isolation — the *block-orbit* $r \geq 2$ *condition*, which denies $\mathcal{A}$ any sub-threshold oracle. The points instantiate Assumption 3: recognizable on their boards but not transmissible without $\theta_{i,j}$ (TKBA₁), and the deep $\theta_{i,j}$ are regenerated only by walking the memory-hard chain (TKBA₂).

The coercion-resistant master secrets are

$$K_i = H(o_i \parallel Sh_i), \quad 1 \leq i, \quad (5)$$

deliberately *cheap*: the orbit yields a nested *family* of master secrets, each a full instantiation of the abstract K of §3, with K_0 set aside since stage 0's board follows from the public salt alone. A memory-hard *last* step would only *prolong* the window in which o_N and the final points are live, worsening the very window (4) shrinks; the resistance therefore lives in *entropy*, standard deep stages carrying 96 bits, so a cheap brute of Sh_N from a bare o_N costs 2^{96} — infeasible by S3 — while the terminal window stays instant.

C.2 Proof of the floor theorem

Theorem 3 is restated from §4: under Assumptions 2–3 and the standard build of (4)–(5), from every seizable state save the residual window of §4.3, a target master secret K_N is prohibitive to reach.

Proof (Proof by case analysis over seizable states). From a seized o_i ($i < N$), reaching K_N requires guessing $Sh_i, \dots, Sh_N$; each Sh_l guess is $2^{r_l \cdot 32}$ and is H^* -gated, since it is checked only by recomputing the memory-hard link (4). Preimage resistance of H, H^* makes the orbit forward-only: a seized o_i cannot yield o_{i-1} . We enumerate:

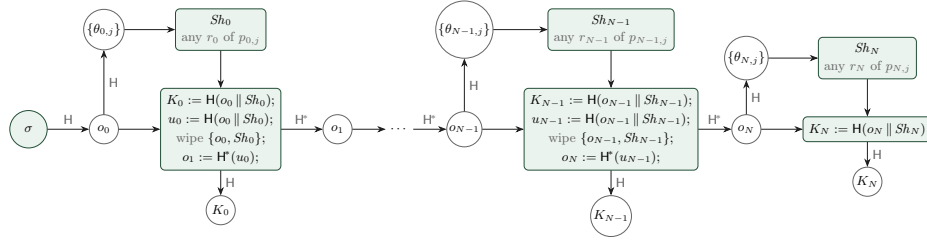


Fig. 2. The TGPO orbit (§4). A public, precomputation-resistant timechain salt σ seeds a chain of N memory-hard stages. At each stage the raw inputs $o_{i-1} \parallel Sh_{i-1}$ are folded by the *cheap* inner hash H and *wiped* before the long memory-hard step H^* , so seizable material exists only for an instant — shrinking the window in which o_{i-1}, Sh_{i-1} are reachable. Each board o_i derives the fractal parameters $\theta_{i,j} = H(o_i \parallel j)$ on which its points $p_{i,j}$ are recognized; any r_i of those points reconstruct the whole Shamir polynomial Sh_i , absorbed into the *next* orbit link, so no single point is checkable in isolation. The terminal secrets $K_i = H(o_i \parallel Sh_i)$ deliberately use the cheap hash, keeping the final window instant. By construction, coercion resistance happens for $i \geq 1$.

- *Pure device seizure* (no coercion; only time-lock-gated [29] blobs at rest). All tacit points are required; K is unreachable.
- *Wrench of stage 0* ($p_{0,\cdot}$, always present since σ is public). $\mathcal{A}$ computes o_1 with one H^* ; from o_1 , K_N costs $2^{\sum_{j \geq 1} r_j \cdot 32}$, which is 2^{96} or more in the recommended standard builds [11], each guess H^* -gated — prohibitive.
- *Mid-orbit seizure* at o_i ($0 < i < N$). K_N costs $\geq 2^{96}$ cheap and is H^* -gated per guess — prohibitive.
- *Residual* o_N present. Treated separately in §4.3.

Every case except the residual leaves K_N at the floor (Stipulation 1).

Remark 18 (Shallow-stage oracle use adds nothing to the floor). The oracle may be exercised on stages sitting below the memory-hard chain — stage 0 above all, whose board $o_0 = H(\sigma)$ follows from the public salt at no derivation cost. *By itself this buys no coercion resistance.* The case analysis already hands $\mathcal{A}$ the shallow points for free (the *wrench of stage 0* case above, and the setting-aside of K_0 in §4): a stage reachable without H^* work contributes no H^* -gated guessing to any path toward K_N , so routing its bits through the perceptual oracle leaves Theorem 3 exactly where it stood. In the ideal model the resistance is carried by the deep stages and by them alone, and nothing below the chain should be counted toward it.

Remark 19 (Redundant oracle use as adversary education). The reference implementation nevertheless routes the shallow stages through the perceptual oracle, and the reason lies not in the ideal model but in the gap between it and a realistic $\mathcal{A}$. Assumption 3 is a claim about $\mathcal{V}$ that a protocol-ignorant $\mathcal{A}$ has no reason to credit on $\mathcal{V}$'s word — disbelief is the ordinary case, and it is what the 2026 streamer incident records [5]. Performing the oracle where it is not needed makes

the ceremony *observable*: $\mathcal{A}$ watches $\mathcal{V}$ recognize points on a board rather than recite anything, and is shown in band — on bits whose exposure costs nothing (Remark 18) — that the remainder of the derivation demands a faculty $\mathcal{V}$ has but cannot hand over. The intended effect is on $\mathcal{A}$'s belief: a coercer who has seen the mechanism work is less prone to read non-transmissibility as refusal, and correspondingly likelier to judge the attempt futile and abandon it.

The force of this is that it is a *demonstration* rather than an assertion. Declarations about the protocol are claims $\mathcal{A}$ may discount at no cost; observed behavior is evidence, and it is cheap to supply precisely because the shallow bits are worthless to protect. It does not, however, discharge NNPP (Assumption 4): $\mathcal{A}$ still cannot verify that no second copy of the secret exists, and no amount of demonstration reduces the kill incentive by proof. What it moves is the probability that $\mathcal{A}$ correctly classifies the situation he is in — a term in his decision, not in the impossibility.

There is nonetheless a second-order effect on that same decision, and it runs in $\mathcal{V}$'s favor: it is the *Obscurity Paradox*, stated and named in Remark 30, where it most sharply separates the non-obscure designs from the obscure ones.

C.3 The residual deadly-race window in full

§4.4 states the criterion $\Delta p_A W > c$ and summarizes why it fails. The five grounds, and the six conditions whose intersection the window occupies, are these.

Two events govern the marginal value of killing $\mathcal{V}$ in this window:

- **Corr** — *the points $\mathcal{V}$ has surrendered so far are correct*; and
- **Esc** — *the ≈ 1 – 2 minutes shaved by brute-forcing the last one or two points, rather than waiting for a cooperating coerced $\mathcal{V}$ to recollect them, make the difference between a successful escape from the crime scene and not.*

Killing can raise p_A only inside $\text{Corr} \cap \text{Esc}$: outside **Esc** the shaved minutes change no outcome, and outside **Corr** (below) killing *lowers* p_A . The five grounds below — grouped by which term of $(1 - p_A) W > c$ they act on — show the inequality fails.

Conditional on Corr, the gain $(1 - p_A) W$ is minimal (p_A is already ≈ 1):

1. a well-trained fractal is recognized in ≈ 1 minute, so brute-forcing the last point saves $\mathcal{A}$ at most that ≈ 1 – 2 minutes over coercing it — bounding the entire interval on which **Esc**, and hence any gain from killing, can act;
2. $\mathcal{A}$'s plan already budgets time-lock or re-derivation, subdue, and interface time, so $\Pr[\text{Esc}]$ is small: the residual minute rarely flips the escape, and killing pays off only inside **Esc**;
3. the terminal K is deliberately cheap ((5) keeps memory-hardness in the orbit, not the terminal step), so $\mathcal{A}$ can *pre-stage* the 2^{32} brute as a “victim-stalls” contingency — recovering the shaved minutes *without* killing, and draining the gain to ≈ 0 .

*Absent Corr, killing makes p_A **fall**, not rise:*

4. if any surrendered point is wrong, $\mathcal{V}$ is $\mathcal{A}$'s only route to a corrected one; eliminating $\mathcal{V}$ burns the coercion retry and collapses p_A toward 0. Since $\mathcal{A}$

cannot verify *Corr* before broadcasting, the *expected* change in p_A from killing is dragged below its *Corr*-only value — and can turn negative.

And the cost c is large:

5. the killing sharply *worsens judicial liability* — murder aggravated by frivolous motive *and* by the victim’s defenselessness, atop armed robbery — so c sits well above the minute-scale $(1 - p_A)W$ the first four grounds leave on the table.

Whether the window opens after the $(r_N - 1)$ -th or $(r_N - 2)$ -th point rests on the softest number of Stipulation 1 — inside the tolerated reasoning either way.

Net. The relevant quantity is *not* $(1 - p_A)W$, which is non-negative by definition ($p_A \in [0, 1]$), but the *utility increment* of the assassinate-then-brute strategy over keeping the victim cooperating. Writing Δp_A for the change in $\mathcal{A}$ ’s win probability that strategy induces, it is rational only if $\Delta p_A W > c$, and

$$\Delta p_A = \underbrace{\Pr[\text{Corr} \cap \text{Esc}]}_{\text{gain (grounds 1–3): minimal}} - \underbrace{\Pr[\neg \text{Corr}] \cdot q}_{\text{loss (ground 4): burned retry}},$$

where $q \in [0, 1]$ is the win probability a still-cooperating victim would restore — via a corrected point — and that killing forfeits. The gain term is small (*Esc* is unlikely, is realized only within *Corr*, and even then is bounded to a minute the brute can be pre-staged around); the loss term is a strict subtraction growing with $\Pr[\neg \text{Corr}]$; and c is large (ground 5). Hence the criterion the strategy must satisfy, $\Delta p_A W - c > 0$, is false with overwhelming probability, so the kill strategy is — much more likely than not — *not* rationally advantageous.

In plain English, the residual deadly-race window lives only in the intersection of a wrench attacker who:

1. is **technically sophisticated** — able to mount the 2^{32} brute and drive the interface. (*Extrinsic to TGPO: granted to $\mathcal{A}$ by Kerckhoffs (Assumption 1), which already hands over the full protocol.*)
2. **landed the attack inside the o_N -live window** — timed precisely enough to catch stage N available, or resourced enough to sustain coercion until o_N was made available;
3. **subdued $\mathcal{V}$ before any reset** of the derivation or the TLP;
4. **judged, right after the second-to-last point, that ceasing the attack immediately was critical to escape** (*Esc*) — despite already having budgeted for a longer encounter (item 2);
5. **could rule out every already-coerced point being incorrect** (*Corr*); and
6. **accepted the liability of a doubly-aggravated murder** — frivolous motive *and* victim defenselessness. (*Once again extrinsic to TGPO: it burdens the deadly-racer attacker of any coercion-resistance protocol.*)

Remark 20 (Cloaking the window). The gain term carries a further hidden factor: the residual is reached only when the attack *coincides* with the brief interval in which o_N is live. That coincidence is itself suppressible. Anonymizing *when* the

vault-opening window occurs — the role of the outsourced anonymous time-lock [42] — drives $\Pr[\text{attack lands in the target } o_N\text{-live window}]$ down, and this factor multiplies the already-minimal $\Pr[\text{Corr} \cap \text{Esc}]$. Here anonymity of the *timing* suppresses a residual that confidentiality of the *secret* alone does not reach.

Remark 21 (Design decision — canonical-island highlighting). Item 5 above (that $\mathcal{A}$ cannot rule out $\neg\text{Corr}$) can be strengthened at the UX layer, to make an *intentionally* wrong prior point plausible. Whenever feasible the interface should keep the canonical-island highlight active: this lets a coerced $\mathcal{V}$ perform a random zoom and then feign a confident, plausible-looking selection of an *incorrect* island. Without the highlight the victim could not reliably produce a plausible selection at all — valid coordinates are a tiny fraction of the space, so an unaided pick would most likely be rejected as invalid — so it is the highlight that makes a deliberately-wrong-yet-credible surrender available to the victim, and thus keeps $\neg\text{Corr}$ live in $\mathcal{A}$ ’s belief (inflating the loss term of Δp_A). A sophisticated $\mathcal{A}$ could force a non-highlighting fork of the app, but that costs further sophistication and wrench-time. This layer sits *on top of* a window of deadly-race vulnerability that is already vanishingly unlikely (the six-way intersection above).

Remark 22 (Feigning is not load-bearing). The $\neg\text{Corr}$ loss term (ground 4) and its UX amplification (canonical- island highlighting, above) only *deepen* the case; they are not what carries it. Grant $\mathcal{A}$ the best of it — $\Pr[\text{Corr}] = 1$, an honest $\mathcal{V}$ in flawless cooperation (D3), with no feigning to fear. Then the loss term vanishes and the kill increment collapses to its gain term alone,

$$\Delta p_A W - c = \Pr[\text{Esc}] W - c,$$

which grounds 1–3 already hold small ($\Pr[\text{Esc}]$ tiny, the shaved interval minute-scale and pre-stageable) against a large c (ground 5). So the residual is bounded on the *gain* side by itself: feigning ($\neg\text{Corr}$) subtracts further and is a welcome bonus, but the window is already small without it — even against a perfectly honest, fully cooperating victim.

Remark 23 (Splitting a stash across several K_i). Nothing forces a holder to route an entire stash through a single master secret; it may be split across several K_i — distinct stages, or independent orbit setups — with two risk-management benefits, each diluting one of the residual failure modes:

1. *Diluted seizure risk* — the proverbial “not all eggs in one basket.” Should the residual *seizure* window (§4.3), or any other failure mode of the construction, expose one K_i , only the fraction of the stash it guards is lost.
2. *Diluted deadly-race payoff.* In the residual *deadly-race* window (§4.4) the kill condition is $(1 - p_A) W > c$; with the stash split, the W at stake in any single window is only a fraction of the whole, lowering $\mathcal{A}$ ’s assassination payoff and making the inequality harder to meet.

D Supplement to the classification

This appendix carries what §5 summarizes: the closure condition on the delegated route, the full self-destruct analysis with the vendor-by-vendor record and the companion decoy, the Obscurity Paradox, and the registry method — the coding tables, the jurisdictional comparison with its confounds, the proxy-victim finding, and the two case analyses.

D.1 When the remote-delegate escape closes

Remark 24 (When the escape closes). The remedy holds only while the delegate is genuinely remote. If the delegate is a friend who happens to be visiting during the attack, *both* problems apply at once — custody was already non-individual, and the racer is now co-located and reachable — so the design becomes deadly-race-susceptible again. The user is thus forced into an awkward corner: trust someone they can rarely be near, and accept a perverse incentive to avoid physical proximity to a trusted friend.

D.2 BitVault

Remark 25 (Sources, and the four columns). *Sources.* No authoritative specification is published. The vendor describes the product as “100% Open Source,” but its code link resolves to an organization with no public repositories (archived at <http://archive.today/2026.09.01-225418/https://github.com/BitVaulty>), and its documentation and audit are served through a gated document-sharing service. What we have is the vendor’s pages, a concept whitepaper [16] that is self-attributed and hosted outside the vendor’s own links, and correspondence of 26–27 August 2026 (all accessed 1 September 2026); they do not agree in every particular. This bounds what can be claimed and is *not* an allegation: we have no evidence that any material was withdrawn, and observed none at any earlier time. The whitepaper’s Taproot/Miniscript policy gates the primary path on `older()` and both fallbacks on `after()`, so the primary delay is a maturation from funding rather than a window on a request, and the multi-year locks do not re-arm.

The columns. Property 1 we leave open: the documentation assumes hardware and offers an optional “copy of A kept in a separate location,” but nothing in the construction forbids running it from memorized seeds and re-deriving on demand, which would be deviceless. Property 2 fails in both directions — the optional off-chain gatekeeper can deny a user the quick path by declining to release the custodian’s co-signature, and past the three-year lock a custodian and that gatekeeper can spend together. Property 4 fails by the two cases above.

Property 3 we also leave open, for a reason worth stating generally: **obscurity does not advertise itself**, so classifying a design as obscurity-dependent is always an inference from what it would have to assume in order to make sense. Here the deadly-race exposure is plain on the published policy, plain enough that a Kerckhoffs-aware design would not ship it; so the design *suggests* an $\mathcal{A}$ assumed

ignorant of it — a reading the scarcity of public-facing documentation would fit. The only other reading we can construct is an $\mathcal{A}$ assumed willing to coerce but not to kill — which is not a milder premise but a wager on his c (Definition 8), staked on the holder’s life. Undetermined is the honest entry between the two, and it costs nothing: the coercion result stands on structure either way.

D.3 The self-destruct class in full

Remark 26 (The obscure self-destruct is the obscure scheme with no winning branch). Obscure defenses are usually compared by *how likely* $\mathcal{A}$ is to be ignorant. The sharper comparison is *how many* independent ignorances each one needs, and here the self-destruct is strictly the worse bet — worse, in particular, than the decoy it is commonly sold beside.

A decoy needs one premise: that $\mathcal{A}$ does not know decoys exist. Grant him that ignorance and the decoy *terminates the encounter* — he leaves with a plausible balance and a story that closes.

A self-destruct needs two, and they are independent. It needs $\mathcal{A}$ ignorant of the feature (sub-Kerckhoffs), so that he reads the failure as breakage rather than as a deliberate move; and it needs him ignorant of NNPP’s consequence (sub-NNPP), so that he does not reason to an undisclosed backup. The second is not implied by the first, and this is the design’s central weakness: *even total success at the first leaves the encounter open*. “This device is broken” does not entail “this person has no bitcoin”; it entails that the instrument for reaching it is broken, whose natural sequel is the question *where is your backup?* — addressed to a $\mathcal{V}$ who is still present and still knows.

So the self-destruct fails against an informed $\mathcal{A}$ for the ordinary Kerckhoffs reason, and against an ignorant one it fails anyway, because the belief it manufactures does not discharge the encounter. It is the one obscure scheme with no branch on which it wins.

We did not have to derive that regress: a vendor documents it. Coldcard ships trick PINs whose actions include a duress wallet — “a personal safety feature. . . If you must reveal a PIN under duress, give the duress PIN instead of your Main PIN” — alongside *Brick Self* and *Wipe Seed* [19]. Its firmware documentation then addresses the case where the decoy is not believed, and its advice is to escalate to destruction: “you may be asked what the actual duress PIN is, while under duress. We suggest providing the ‘brickme’ PIN in that case” [20]. That is the manufacturer contemplating an $\mathcal{A}$ who knows decoys exist, and recommending, as the answer, an irreversible erasure performed with $\mathcal{V}$ still in the room and still able to be asked for a backup — the ignorance-count argument above, arrived at independently and published as guidance. The same document instructs the holder to “triple-check your backups are correct” before enabling that PIN, so the backup presupposition of §5.3 is not peculiar to one vendor either.

Remark 27 (The class, and what each vendor actually claims for it). Three vendors ship the pair, which is what makes this a design consensus rather than one

firm’s misstep; but they do not make the same claims, and the difference is worth recording precisely. Blockstream markets the *erasure* for physical threat while claiming nothing of the kind for its passphrase wallets. Coldcard markets *both* halves, describing the duress wallet as a safety feature and the brick PIN as the fallback when it fails. Trezor is the mirror image of Blockstream: its passphrase wallets are sold on deniability, while its wipe code — which the vendor itself calls a “self-destruct” PIN, and which erases the recovery seed with it — is documented without any mention of duress, coercion or physical threat [43]. That vendor was moreover asked, in January 2022, to make the wipe *concealable*: a user-defined message on entry, so that an erasure could be presented as a hardware fault rather than announcing itself. The request was closed as *not planned*, and the device still reports the wipe plainly [44]. Component (b) is therefore not intrinsic to the self-destruct class at all: it is a presentation one vendor builds and another, asked for it directly, refused. What §5.3 analyses under (b) is specific to the “Internal Error” design, and a wipe that announces itself carries (a) alone — which, by the argument above, is no coercion resistance at all, and is not sold as any. Two of the three therefore decline to make the coercion claim on one of their two mechanisms. We note it because the criterion is not advanced against vendors, and a vendor that ships a mechanism without claiming it defeats coercion has not made the error this section is about.

Remark 28 (Two bluffs the erasure invites). An $\mathcal{A}$ who knows the design (Assumption 2) need not walk into the erasure at all. Two alternatives stand open to him, each a wager rather than a certainty:

1. *Take the device hostage up front.* Rather than enter any PIN, seize the device and hold it as a hostage token (§3.7, Remark 14). An erase PIN is irrelevant to a coercer who never authenticates; and by Remark 15 the terms on which $\mathcal{V}$ ransoms the device are themselves a probe for the very backups $\mathcal{V}$ cannot disprove.
2. *Coerce against deployment.* Brutalize $\mathcal{V}$ to deter the PIN’s use in the first place, wagering that a holder under sufficient duress will not spend the one irreversible move available to them.

Neither wager is sure to pay, and that is precisely the point: both are bluff games, and the holder who has bet on an erasure has entered one without holding the means to end it. A $\mathcal{V}$ who erases and then *truthfully* denies retaining any backup arrives at the position the registry already records — the streamer whose denial was simply disbelieved, and who was tortured for hours regardless [5] (§5.5).

The “Internal Error” disguise inherits the same objection from a different direction: it is a bet on $\mathcal{A}$ ’s ignorance of a consumer product, and it degrades exactly as that product’s own public documentation becomes known (§3.6). The class therefore lands in the **naïve** tier of Definition 12 precisely as that definition describes it — an obscure self-destruct, safe only against a sub-Kerckhoffs attacker — and it is worth keeping the self-destruct distinct from the *cancel* key named alongside it there. A cancel reverses a pending action and thereby names a second racer; a self-destruct destroys a path and names none. Only the former

is subject to Lemma 1, which is why the two fail differently despite sharing a tier. Remark 13 has already disposed of the tier as a whole, by the dilemma that an $\mathcal{A}$ whose ignorance can be assumed makes the elaborate mechanism unnecessary, and one whose ignorance cannot makes it ineffective; the present subsection is that dilemma instantiated in a shipping product.

The companion decoy, classified separately. The same device ships a second and quite different obscure mechanism, and the two should not be run together. A BIP-39 passphrase “is added to your Jade’s recovery phrase or SeedQR and is used to create your wallet”; “any number of passphrases can be entered to change the wallet you will log into”; and the device “will not store your passphrase” [45]. Distinct passphrases therefore yield distinct wallets from one recovery phrase, and the familiar community practice is to keep a low-balance wallet at the empty passphrase to be surrendered under coercion.

Two observations. First, and to the vendor’s credit, *Blockstream does not market it this way*: the passphrase documentation claims no deniability, decoy or duress use, in pointed contrast to the erase-PIN pages, which name physical threat explicitly. The duress framing here is the community’s, not the manufacturer’s, and the distinction is worth preserving in any account of who claimed what.

Second, the mechanism is the plausible-deniability layer already treated in §3.6 and dismissed for the same Kerckhoffs reason: it rests on $\mathcal{A}$ ’s ignorance, and an $\mathcal{A}$ who knows that passphrase wallets exist simply persists past the surrendered one, with NNPP guaranteeing $\mathcal{V}$ cannot close the question. It is not, however, the *same* failure as the self-destruct’s, and by Remark 26 it is the less bad of the two: the decoy at least has a branch on which it wins, since an $\mathcal{A}$ who believes it has been paid and can leave. The self-destruct has none. That a single product ships both, marketing only the weaker one for physical threat, is the clearest illustration in this section of how little the design space has been mapped against the Criterion.

Remark 29 (The decoy is sold, and coached for credibility). The decoy is not merely a practice the community tolerates; it is a product and a curriculum. A Spanish-language self-custody consultancy lists, among paid service tiers, “*Passphrase (tu bóveda señuelo)* — Billetera señuelo + billetera real. Protección ante coacción/coerción” — a decoy wallet marketed, in as many words, as protection against coercion [46]. Published guidance goes further and coaches the performance: one guide has the holder keep a small balance on the bare seed as “what you’d hand over under physical coercion”, and adds that “for this to work, the decoy balance must be *believable* — enough to look like a real wallet, not enough to devastate you if lost” [47].

That last instruction is the naive tier stated as a training objective. A decoy need be believable only to an $\mathcal{A}$ who might disbelieve it, and the counsel is silent on what happens when he does — which, by Definition 9, is the whole question. A better-rehearsed lie raises the chance $\mathcal{A}$ takes the bait at the margin; it does not create a branch on which the scheme wins against an $\mathcal{A}$ who knows decoys exist, and NNPP (Assumption 4) guarantees $\mathcal{V}$ cannot close that question however well

the performance goes. What is sold as a mechanism is a wager on the attacker’s ignorance, priced and taught as though it were a control.

The counsel is also self-defeating in aggregate. Publicised training raises the *prior* that a fluent denial was rehearsed: once $\mathcal{A}$ knows credibility is taught and sold, fluency stops being evidence of truth and becomes evidence of coaching. The remedy degrades the thing it sells — and not only for its customers. A holder who never trained now denies into an $\mathcal{A}$ who discounts fluent denial generally, so the cost falls on people who bought nothing. This is the Obscurity Paradox (Remark 30) run backwards: there, a public non-obscure design makes $\mathcal{V}$ ’s claims *more* plausible; here, a public obscure training makes them *less*.

Remark 30 (The Obscurity Paradox — non-obscure designs strengthen obscure ones). TGPO enters neither bluff. There is nothing to erase, the deep secret having never been on the device (Theorem 3), and so nothing whose non-deployment can be coerced. But the comparison yields more than the absence of a failure mode, and the surplus runs in a direction the obscurity argument would not predict.

An obscure defense asks $\mathcal{A}$ to believe a *claim* — that this is the only wallet, that nothing further remains — against both an incentive to disbelieve it and, by NNPP, no means of settling it. A non-obscure defense changes what making that claim costs, and it does so by the mechanism set out in Remark 19: an $\mathcal{A}$ who credits that an effective mechanism is in place is crediting that lying is not $\mathcal{V}$ ’s only available tool, and a holder with effective alternatives has correspondingly less reason to resort to one.

The ordering this inverts is worth stating plainly, and we name it **the Obscurity Paradox**: *a publicly specified, Kerckhoffs-clean design improves the standing of precisely those obscure denials it was built to make unnecessary*. Obscurity remains unsound as a primary defense, for the reasons §3.6 gives. It is not, however, indifferent which design it accompanies — a denial made from within a mechanism that removes the need to lie is a stronger denial than the same words spoken from within one that requires it. The same asymmetry has an epistemic cost, taken up next (Remark 31).

Remark 31 (Obscurity anosognosia — why the crisis resists diagnosis). Two censoring mechanisms compound, and between them they remove the evidence that would settle whether the Lemma’s prediction is being enacted in the field.

The first is the one this appendix has already named: obscurity does not advertise itself. No vendor documents a scheme as depending on $\mathcal{A}$ ’s ignorance, so for any given victim the question *was this holder running an obscurity-dependent design?* has no recorded answer. The exposure variable is not merely unmeasured but unmeasurable from public sources, which forecloses the obvious epidemiological question — what do the victims of prolonged coercion, or of the terminal move, have in common?

The second is recorded in §5.5: a homicide is reported as a homicide, not as a robbery whose victim was then killed, so the enactments most diagnostic of the Lemma are the least likely to enter the registry at all. The numerator

is censored by how the terminal event is classified; the covariate is censored by what vendors do not say.

The structure is that of *anosognosia* — the dementia patient whose disease erodes memory and, with it, the capacity to notice that memory is eroding. One cause, two effects, and the second conceals the first. Obscurity is simultaneously what makes a design fail under Kerckhoffs and what erases the record by which the failure would be recognised. We draw no empirical claim from this: we cannot, which is the point. What it establishes is that the diagnosis is not available from the data by waiting, and that a field which waits for it will keep not finding what it is not positioned to see. That is an argument for the formal route — the bar has to be derived, because it will not be measured.

D.4 Registry method and coding tables

§5.5 reports the headline findings of the coding. The method, the full tables and the caveats are these.

Short attacks dominate; the registry over-counts the dramatic. Most wrench attacks are *hand-over-now* events — a threat, an immediate transfer, minutes on scene; the long, cinematic kidnapping is the exception. A media-sourced registry inverts this, because a mutilation-kidnapping is news where a knife-point mugging is not, so the kidnap share coded below *over*-states the long tail rather than establishing it. That TKBA is realistically *less* effective against a long-duration attack — where hours of pen-and-paper effort under coercion *might* walk a deep stage — is granted, but it is *speculative* pending the field study of §7. What is *not* speculative is that the fielded alternatives fail; the cases below make that concrete.

Signal (headline keyword coding)	Incidents	Share
Killed / murdered victim (<i>a lower bound</i>)	18	5.1%
Kidnap / hostage / ransom (long-hold)	139	39.5%
Armed robbery at gunpoint (short-hold)	101	28.7%
Home invasion / break-in / raid	40	11.4%
Torture or physical violence	82	23.3%

Categories overlap (a kidnapping may also be violent or fatal) and do not sum to 100%; shares are of all 352 incidents, and the modality labels are a headline proxy, not a measured duration. The sample is media-reported and survivorship-biased — deterred, unreported, and silently-fatal cases are absent — so every figure is directional and the fatality share in particular is a floor. Remark 31 gives the two mechanisms that censor it, and why they leave the diagnosis unavailable rather than merely imprecise.

The mix is not universal, and the Lemma’s bite tracks it. The global shares above average over jurisdictions whose compositions differ by an order of magnitude. Taking the ratio of long-hold to short-hold codings, $K:G$:

Jurisdiction	n	Kidnap	Gunpoint	$K:G$
Brazil (<i>small n</i>)	12	75.0%	8.3%	9.0
France	61	57.4%	8.2%	7.0
<i>All incidents</i>	352	39.5%	28.7%	1.4
United States	59	28.8%	40.7%	0.7
Russia (<i>small n</i>)	10	30.0%	50.0%	0.6

The comparison that carries weight is France against the United States: the two largest national subsets, of near-identical size (61 and 59), with *inverted* compositions — a tenfold difference in $K:G$. This is the sharpest available check on the media-inflation caveat just stated. That caveat explains a *level* — why the registry over-counts dramatic incidents everywhere — but it cannot explain this *ordering* without the further claim that French reporting is ten times more kidnap-selective than American, in a market that is at least as heavily covered. The parsimonious reading is that modality composition genuinely varies with local conditions — firearm availability, the presence of organized kidnap-for-ransom capability — as it does for kidnapping outside this domain entirely.

Two confounds remain and are not dismissed. Outside France and the United States the subsets are too small to carry an argument, so the other rows are directional only. And the coding is applied to English-language headlines, several of them translations, so a systematic vocabulary difference — French sources reaching for “kidnapped” where American ones reach for “robbed at gunpoint” — would produce part of this spread without any difference in the underlying events.

If the composition is real, it matters for scope rather than for validity. The Deadly-Race Lemma (Lemma 1) is a statement about the *released* holder, and release is the trigger (Remark 1); a hand-over-now robbery at gunpoint approximates the clean $p_A = 1$ endpoint — the attacker takes what is reachable and leaves — whereas a multi-hour hold ending in a deliberate release is exactly the gray outcome. Coercion risk is therefore not merely higher in some jurisdictions but *differently shaped*, and the No-Gray-Area Criterion (Criterion 1) does its work where long holds dominate. That is the regime the record is currently growing into: France alone accounts for a majority of the incidents recorded in 2026.

D.5 The proxy-victim finding

The attacker often does not need the holder at all. A fourth signal falls outside the keyword table because it was coded by hand: in 12 of the 352 incidents (3.4%) the person seized was *not* a holder but a relative held to compel one — a parent, a spouse, an adult child.⁴ Two features of this subset matter more than its size.

⁴ Keyword screening surfaced 16 candidates; each was read individually and four were discarded as false positives, including one in which “son of” named a *perpetrator* rather than a victim. **registry-analysis/** carries the screen and the full audited

It is *recent and concentrating*: eleven of the twelve fall in 2023 or later, and *nine* occur in a single jurisdiction (France) whose recorded incidents rose from about one a year through 2024 to 22 in 2025 and 34 by August 2026. Whatever share of that rise is genuine incidence rather than reporting intensity, the tactic is not a historical scatter; it is being adopted.

It is *not* the hostage token of §3.7, and the difference matters. There the leverage is a *seizable object* the protocol requires, which devicelessness removes by having no such object; here it is a *person*, whom no custody design renders unseizable. What this subset shows in the field is clause D8 failing: such an attacker needs no cryptographic access and no cooperation a protocol could withhold — only leverage, and a relative supplies it. This bears directly on the delegated route (R2), whose clearance rests on the deciding party lying beyond $\mathcal{A}$'s coercive reach. The delegate a holder is most likely to actually appoint — a spouse, a parent, an adult child — is precisely the party this subset shows being seized. The route is not thereby refuted: it is bounded. A delegate who is reachable is not a second racer but a second victim, so R2 clears the bar only for delegates whose unreachability is structural rather than assumed — and that requirement, not the arrangement's convenience, is what the data prices.

D.6 Two cases in point

A case in point — Balland. The January 2025 abduction of David Balland, a co-founder of the hardware-wallet maker Ledger [23], is instructive precisely because it is an *embarrassment to the self-custody creed*. A professional champion of self-custody was seized from his home and mutilated — a finger severed to force payment — and neither wealth nor prominence prevented it. He was freed by an elite *state* police unit, and the ransom — paid not by Balland but by an associate — was clawed back largely because part of it moved in a *centralized* stablecoin whose issuer could freeze it. A self-custody pioneer made whole by the state and by centralization: two mechanisms external to, and in tension with, the creed. Only the issuer freeze is independently verifiable; the remainder rests on the parties' accounts, according to which roughly 5% was never recovered. The case also indicts the delegated route (R2): an associate *did* pay, which is exactly the compliance clause D8 assumes away — a real instance of the indirect coercion, routed through the victim's body to a third party who will not refuse. For a brief hand-over-now robbery a remote delegate is unreachable in time and so seems to help; but merely *not carrying* the device — leaving the wallet in a home vault — does the same, while for the long attack the delegate is not out of reach but is the ransom channel itself. The lesson is not that Balland erred: it is that, against a determined long-hold attacker, the fielded options collapse to state rescue and centralized freezing, whereas the tacit route removes the

list. Because the coding reads a one-line headline, an incident that does not name a relative is no evidence that none was involved: 3.4% is a floor, and the coding can only understate it.

leverage — no standing device, no hostage token, no delegate to pay — rather than relying on ex-post recovery.

A case in point — RATIRL. The registry coded above records the obscurity route’s failure directly. In one of its 2026 incidents (Sweden, 13 January) [5] a League of Legends streamer was bound and tortured for hours by armed intruders demanding cryptocurrency he denied holding; unable to credit the denial, they persisted long past any value the robbery could rationally return, leaving finally with only his Pokémon-card collection. This is Corollary 1 in the field: where the community norm is “deny that you own Bitcoin,” a coerced denial is *pre-discredited* — presumed to be the denial-script — so a Kerckhoffs-educated $\mathcal{A}$ gets no exculpatory update, and the rational continuation of an interrogation that opens with a denial is *more* coercion, not less. The case fixes the *sign*: obscurity does not merely fail to protect — it can convert a robbery into a torture — *negative* coercion resistance, not a mere absence. And it does so on *every* reading of the denial, which forks three ways — each indicting obscurity. **(i)** If the denial was a defensive lie (there was a stash), secrecy still did not avert the assault — a textbook Kerckhoffs failure — and his only residual recourse is to *double down* on it and wager against a repeat. **(ii)** If it was true (no stash), he bore obscurity’s *negative externality*: a genuine non-holder attacked only because real holders are coached to look exactly like him — the very coaching that collapses the statistical holder/non-holder distinction. **(iii)** If the truth is in between — a real but smaller-than-presumed stash, too small to be worth the judicial exposure the attack incurred — *both* pathologies operate at once: others’ obscurity redirected the attack onto him, and his own failed to stop it. That all three are simultaneously plausible is itself the verdict: in the aggregate, across a growing population of holders and lookalikes, each is already occurring — and the obscurity culture, exactly as Kerckhoffs predicts, prevents none of them.

E Notation, and two refinements of the model

E.1 Notation

The symbols below recur throughout; each is defined in full where it is introduced, and all are gathered here for reference.

Principals, assets, economics

$\mathcal{V}, \mathcal{A}$	holder (victim) and coercer (attacker)
K	master secret gating the asset
W	value of the asset to $\mathcal{V}$
c	expected disutility to $\mathcal{A}$ of killing $\mathcal{V}$
π	$\mathcal{V}$'s <i>pivotality</i> — win-probability $\mathcal{A}$ gains by removing $\mathcal{V}$

Adversary, game, safety

WDY	the Wrench–Dolev–Yao adversary, fixed by clauses D0–D8 (Assumption 2)
p_A	probability $\mathcal{A}$ reaches K after the encounter; <i>gray</i> iff $0 < p_A < 1$
t	coercion-time budget (how long $\mathcal{A}$ can sustain coercion)
G	protocol time-gate (enforced latency to complete the derivation)
Corr	residual-window event: the points $\mathcal{V}$ has surrendered so far are correct (§4.4)
Esc	residual-window event: the ≈ 1 –2 min shaved by brute-forcing the last points decides escape from the scene (§4.4)

Construction (TGPO orbit §4)

σ	public, precomputation-resistant timechain salt
H, H^*	unspecified cheap hash; unspecified memory- and depth-hard (Argon2-class) hash
o_i	orbit state at stage $i \in \{0, \dots, N\}$; $o_i = H^*(u_{i-1})$ (with $o_0 = H(\sigma)$); $N+1$ stages total
N	number of <i>deep</i> stages — those requiring at least one memory- and depth-hard hash (H^*) to derive
Sh_i	Shamir polynomial at stage i ; r_i of its points reconstruct it
u_i	$H(o_i \ Sh_i)$: commitment of the wiped values o_i and Sh_i
K_i	$H(o_i \ Sh_i)$: master secret correspondent to stage i .
$p_{i,j}, \theta_{i,j}$	point j of stage i ; parameters for fractal j of stage i
r_i	Shamir threshold at stage i ($r_i \geq 2$)
64^*	2^{64} memory- and depth-hard evaluations (infeasible)
floor	<i>prohibitive</i> cost: $\geq 64^*$ hard <i>or</i> $\geq 2^{96}$ cheap (Stipulation 1)

E.2 The two tacit-seizure clauses

Remark 32 (The two non-seizure clauses coincide for design). For protocol-design purposes the tacitness of the entropy-carrying secrets (D6) and that of the perceptual-oracle parameters (D7) are logically equivalent. If the parameters were transmissible, a wrench attack would seize them — and, available within that *same* encounter, they would then serve to seize the entropy-carrying points themselves; so non-tacit parameters force non-tacit points. Conversely, if the points were somehow transmissible without the parameters ever being available, then the tacitness of the parameters is irrelevant, the points having already

fallen. Either way the binary question “does a single seizure reach K ?” turns on the two clauses together, and the model may treat them as one.

Remark 33 (Where the distinction still matters). The distinction can nonetheless matter for a *quantitative* refinement — a statistical distribution of the wall-clock needed to coerce either (up to sufficient number of most-significant bits) given $\mathcal{A}$ ’s coercion time budget t (Definition 8). Still, a secret that is *theoretically* transmissible, but only with pen and paper, an adversarial fork of the application, and long hours to days of focused mental effort under perfect $\mathcal{A}$ – $\mathcal{V}$ cooperation, is essentially as good as ideal. Until empirical studies of TKBA (Assumption 3) are done, the actual relevance of that nuance is purely speculative.

Remark 34 (TKBA restates the tacit-seizure clauses). Assumption 3 is not an independent hypothesis: it is clauses D6–D7 lifted out of the adversary model and stated as a premise on the secret. The clauses say $\mathcal{A}$ cannot seize the tacit points or the perceptual parameters even under full cooperation (D3); TKBA says the same thing about the secret itself — recognizable but not transmissible — with no adversary mentioned. They are two faces of one assumption (exactly as D0 and Assumption 1 are two faces of Kerckhoffs), and by Remark 32 the two clauses are themselves one. We keep both forms because each is the natural one in its place — the premise where the construction is motivated (§4), the adversary clause where seizure is reasoned about.

F Supplement to the quantum note and the limitations

F.1 Grover against the cheap bound

Remark 35 (The one bound Grover does dent: the live- o_N residual). The duality’s *cheap* bound is a different matter. The floor also clears 64^* via a cheap 2^{96} ledger brute of Sh_N (Stipulation 1, S3), and that cheap side inherits *no* depth-hardness to guard it: Grover halves it outright to $\sim 2^{48}$. Concretely, a Quantum WDY attacker landing on an imminently- or currently-live final board o_N — the same coincidence that opens the tolerated residual window (§4.3) — faces only the cheap 2^{96} brute of the terminal secret, and Grover reduces that to a $\sim 2^{48}$ (“48-qubit”) floor.⁵ This touches nothing but the residual: it does *not* reach the memory-hard 64^* bound (guarded above), and it applies only inside the already- ε -bounded live- o_N sliver, whose incentive-safety argument (§4.4) turns on the kill economics, not on the brute’s width — a 2^{48} Grover search buys $\mathcal{A}$ nothing a classical 2^{32} brute in that same window does not already grant.

⁵ Each Grover attempt is one cheap evaluation of the terminal step — a single $H(o_N \parallel Sh_N)$ key-derivation followed by the ledger decryption/spend check — and *nothing* more: the search is over the 96-bit Sh_N *directly*, so Shamir reconstruction (the holder’s step of interpolating Sh_N from r_N recognized points) never enters the oracle loop. One quantum attempt is thus KDF + decryption, not Shamir + KDF + decryption.

F.2 Memorization cost and the adoption precedent

Memorization cost, and why it is affordable. A separate objection to a brain-only setup is not whether the oracle *exists* but whether living with it is too costly — carrying ≥ 96 bits of tacit recall indefinitely. Two things bound that cost. First, retention is *assisted*: a spaced-repetition companion tool [36] schedules review (an SM-2 / Anki-class algorithm), turning a one-off feat of memory into a light, decaying maintenance load. Second, the feat is historically ordinary: before mobile computation offloaded it, ordinary people routinely held a handful of phone numbers, several street addresses, and a working mental map of their city’s grid — recall comparable to what TKBA asks. The cost is real but bounded, and it is the price of a secret no device can be seized to reveal.

Adoption precedent — demanding standards clear once the pain is legible. The cost objection also has an empirical form — *users will never accept this much ceremony* — and it met a natural experiment in the July 2026 seed-entropy incident (§8). Per the vendor’s own advisory, seeds created with 50-plus fair, independent dice rolls were not at risk from the defect [37]; within days, community guidance converged on exactly that manifestly labor-intensive procedure as the default counsel. Effort tolerance, in other words, is not a constant of the population; it reprices overnight once a threat is legible. The wrench-attack class already shows the same willingness in a smaller way, and spends it badly: decoy construction is sold as a paid service tier and published guidance coaches the performance (Remark 29). The effort budget exists; what it currently buys is a scheme the Criterion rejects. TKBA’s memorization load stands to the wrench-attack class as dice-rolling stands to entropy failure — demanding, verifiable by the holder alone, and adoptable at scale when the danger is understood — and making the danger legible *before* the corresponding incident is precisely what a formal bar is for.

G The discretion counsel, in full

§3 and §5 treat obscurity as a defense. The prior question — whether the records that make a holder a target need exist at all — is raised by the counsel to buy only outside the identity-verified channels and leave nothing fiscally reportable behind. That this lowers exposure we grant, as legitimate personal hygiene. The introduction’s footnote names the three things that nonetheless keep it from working as a security mechanism; here they are in full.

First, exposure ratchets. A record cannot be uncreated and a leak cannot be un-leaked, so the strategy pays only *ex ante*, and only for the holder whose data has never once escaped; for everyone else there is no longer anything protectable to protect. Worse, the two curves run against the holder: the cost of maintaining discretion compounds with every year held and every counterparty met, while its marginal return falls as the stock of records already leaked grows. *Second, for the compliant holder the strategy is not lawfully available.* Where holdings or gains are legally reportable, leaving no fiscal record is not discretion but non-declaration — tax evasion, a criminal offense — so the counsel reduces to an

instruction the law-abiding holder cannot follow. Nor is the resulting record safe by virtue of sitting with the state: in France, the jurisdiction carrying the largest share of recorded incidents, an employee of the tax administration was placed under judicial investigation in 2025 for allegedly consulting the authority’s internal database and passing the results to organized crime, with investigators reporting queries aimed specifically at cryptocurrency investors [48]. The compliant holder is thus asked to choose between an offense and a database that has been mined for targets. *Third, and decisive here*, discretion is not a *mechanism* at all: under Assumption 1 the adversary is granted the holder’s situation, so any defense contingent on their ignorance is out of scope by construction.